

INTRODUCTION

1.1 ABOUT THE PROJECT

The Judicial Management System has been continually enhanced since the software was developed. Each module of the system will work hand-in-hand with the other modules to create an electronic flow of case related information. The electronic flow of case related information from one judicial office to another reduces redundant data entry and/or manual record keeping processes. It also reduces the possibility of paper related errors and speeds the processing of information within the Judicial System. The following is an example of how our system works: When dispatch receives a call and dispatches an officer to the scene of a reported crime, the dispatcher may begin a computer case record of the offense. When the Criminal Investigation Division of the Sheriff's Department receives the file, all information is input during the initial investigation, or booking of suspects, is available on-line. Any previous information on suspects arrested or investigated by the county will also be available on-line. When this case is presented to the District Attorney's Office they will review the information on-line, adding anything necessary for their office. Once prepared for prosecution, the necessary forms for processing the case are printed using the information accumulated in the system. Confidential information is secured via a series of security levels so only information pertinent to an employee's job may be accessed by that employee's password. When the forms generated by the prosecuting attorney are filed with the clerk's office, the clerk enters the case number, necessary codes for report generation and other information necessary for their office while confirming the accuracy of the existing case information. The Court Coordinators then utilize the system and accumulated information for generating court dockets. Upon completion of the case, the disposition of the case may be entered during court as rendered by the Judge. Court documents are printed and all necessary information is now available for the Probation Office.

1.2 ORGANISATION OVERVIEW

Bethel Computers Malayalam is a distinguished educational institution, highly regarded for its commitment to providing top-tier computer education. As an affiliate of C-dit (Centre for Development of Imaging Technology), a Government of Kerala initiative, Bethel Computers Malayalam stands as a key player in the realm of skill development and digital literacy. They offer a diverse range of computer courses, catering to both beginners and advanced learners, and are known for their comprehensive curriculum that equips students with the necessary expertise to thrive in today's technology-driven world.

The institution has established a solid reputation in the educational sector due to its unwavering focus on delivering quality education. Over the years, it has built a loyal student base, with a remarkable track record of 100% client retention. This level of loyalty speaks volumes about the effectiveness and relevance of the programs offered, as well as the institution's ability to foster a conducive learning environment that meets the evolving needs of the digital age. The core of Bethel Computers Malayalam's success is its commitment to quality. The institution recognizes that true quality requires more than just knowledge—it demands discipline, hard work, creativity, and innovation. Their educators and staff follow a work methodology that is both rigorous and flexible, ensuring that students receive not just theoretical knowledge but also practical, hands-on experience that is crucial for real-world application.

SYSTEM ANALYSIS

INTRODUCTION TO SYSTEM ANALYSIS

System analysis is a structured approach to understanding, designing, and improving systems to meet specific goals. It combines principles from various fields such as computer science, engineering, and management to break down complex systems into smaller, manageable components. By analysing the system's structure, behaviour, and interactions, analysts can develop solutions that optimize performance and minimize risks.

The Judicial Management System (JMS) is designed to enhance case management and streamline workflows within the judicial system. It automates tasks, such as case tracking, document storage, and scheduling, which reduces the administrative burden and the likelihood of manual errors. The system also facilitates better communication and collaboration between judges, court staff, and attorneys, ensuring smoother information sharing and case processing. By digitizing the judicial process, the JMS increases efficiency, reduces costs, and improves transparency. However, its implementation may face challenges such as high initial costs, resistance to change, and potential technological barriers, all of which need to be addressed for the system to be fully effective. Despite these challenges, the JMS has the potential to significantly improve the speed and quality of the judicial process while ensuring better data security and accountability.

2.1 EXISTING SYSTEM

The current judicial system faces numerous challenges that undermine its effectiveness and accessibility for the general public. One of the major issues is the limited interaction between the judiciary and society, which creates a significant disconnect between citizens and the legal system. This detachment often results in a sense of alienation, where individuals feel their concerns are not adequately addressed or understood by the legal system. In addition, judicial procedures are frequently slow and inefficient, with cases often taking years to resolve. This backlog leads to considerable delays, causing frustration and dissatisfaction for all involved parties, including plaintiffs, defendants, and legal professionals. In some instances, clients struggle to keep track of their cases, adding to the confusion and uncertainty that already surrounds legal proceedings.

Another significant issue is the cumbersome and complex verification process for legal documents and judicial actions, which further complicates the experience for litigants trying to navigate the system. The time-consuming nature of these processes increases the frustration of those seeking justice, especially when faced with the intricate bureaucratic layers that seem to slow down progress. Moreover, the high costs of litigation, including expensive legal fees and court charges, create barriers to access, particularly for low-income and marginalized communities. As a result, these individuals often face an insurmountable obstacle in seeking legal recourse, further entrenching inequality within the justice system. These barriers highlight the urgent need for judicial reform to simplify processes, reduce delays, and make legal services more affordable. By addressing these challenges, the judicial system can enhance its efficiency, accessibility, and fairness, ultimately fostering greater public trust and ensuring that justice is more readily available to all citizens, regardless of their background or resources.

2.1.1 Limitations of Existing System

- **Lack of Interaction Between Legal Entities:** Different departments within the judiciary work in isolation, causing inefficiency and delays in case processing. Data sharing becomes time-consuming and prone to errors.
- **Delays in Legal Procedures:** Heavy paperwork, prolonged verification, and bureaucratic obstacles lead to delayed case dispositions, causing frustration for litigants and legal professionals.
- **Outdated Record-Keeping:** Litigants and lawyers struggle to track case statuses due to outdated record-keeping procedures, leading to confusion and a perception of inefficiency.
- **High Legal Costs:** Legal services, court fees, and document certification make justice unaffordable, particularly for vulnerable groups, discouraging people from seeking legal recourse.
- **Complex Verification Processes:** Verification and documentation procedures are needlessly complex and cyclical, causing delays and errors in case processing.
- **Risk to Judicial Records:** Paper-based systems and lack of digital infrastructure put judicial records at risk of loss, tampering, or unauthorized access, compromising case confidentiality.
- **Limited Public Engagement:** The judiciary has minimal interaction with the public, leading to a lack of awareness, trust, and involvement in legal processes.

2.2 PROPOSED SYSTEM

The envisioned system intends to completely automate the judiciary, including courts, judges, attorneys, and citizens. The innovation is meant to improve both administrative functions as well as case management, providing an end-to-end solution suited to Judiciary, Court Administration, Case Management, and Court Reporting Services. With the adoption of contemporary technology, the system aims to simplify judicial processes, increase efficiency, and achieve a more responsive legal system.

One of the most notable aspects of this automated system is its friendly interface that is accommodated for all stakeholders irrespective of their technical background. The intuitive interface is meant to help make legal procedures easy to use and also minimize the learning curve for all levels of users. Aside from usability, the system also minimizes procedural delays, where cases could be processed and settled at a much quicker pace. By minimizing inefficiencies, it also reduces the overall expense of legal proceedings, making justice more accessible to individuals and families. This modernization initiative not only enhances the speed and efficiency of judicial processes but also amplifies public access to justice, making the legal system more inclusive.

The system also has an online case registration component, where users can start and follow up on their cases remotely. This improves accessibility and provides a way for people to interact with the judicial system without the inconvenience or physical travel to court offices. Further, the system simplifies police verification procedures, minimizing the time spent on conducting background checks and validation of cases. This not only accelerates the judicial process but also releases the law enforcement agencies from the administrative load, allowing them to carry out more important functions. Overall, the plan of an automated judiciary is a revolutionary solution to the endemic problems of India's judiciary.

2.2.1 Advantages of proposed system

- **Enhanced Accessibility** – People can register and track cases over the internet from anywhere, reducing the need to physically visit court offices.

- **Faster Processing of Cases** – Automation minimizes procedural lags, and the cases are processed faster.
- **User-Friendly Interface** – Simple and intuitive to access by all stakeholders regardless of technical expertise.
- **Lower Legal Costs** – Effective processes reduce administrative costs, and justice costs less for citizens and families.
- **Efficient Police Verification** – Simplified and speeded-up verification procedures lighten the load on police organizations.
- **Increased Transparency** – Litigants, attorneys, and other parties can easily obtain information about cases, increasing transparency in court proceedings.
- **Enhanced Coordination Among Judicial Organizations** – Coordination among courts, prosecutors, police agencies, and other departments allows for easy sharing of information.
- **Less Scalability for Future Growth** – The system is scalable and expandable with increasing legal requirements and case volumes.
- **Less Court Staff Burden** – Automation reduces tedious administrative tasks, allowing judicial staff to focus on critical tasks.

Through the elimination of inefficiencies in the current judicial system, the new system redefines the legal environment so that speedy, equitable, and affordable justice is made available to everyone.

2.2.2 Features of Proposed System

The proposed system offers a range of features aimed at revolutionizing the agricultural marketplace and enhancing the experience for both farmers and buyers:

- **Lower Legal Costs** – Effective processes reduce administrative costs, and justice costs less for citizens and families.
- **Automated Case Management**: Streamlines case processing, significantly reducing procedural time and enabling faster case resolutions.

- Online Case Registration: Users can initiate and monitor cases from any location, providing greater flexibility and access.
- Reduced Legal Costs: Automation lowers overall costs, making the legal process more affordable for individuals and families.
- Police Verification Integration: Simplifies and speeds up the police verification process, reducing time spent on procedural tasks.
- Centralized Data Access: All case-related data is stored digitally, allowing easy access and better management for courts, lawyers, and petitioners.
- Seamless Communication: Facilitates communication between petitioners, lawyers, and judges through the system, improving collaboration and transparency.
- Enhanced Security: Digital records and secure systems ensure case confidentiality and data integrity.
- Greater Public Access: Increases access to legal services by enabling citizens to interact with the judicial system remotely and efficiently
- User-Friendly Interface: With an intuitive and easy-to-navigate interface, the system caters to users of all levels of technological proficiency, ensuring a smooth and hasslefree experience for both farmers and buyers.

2.3 REQUIREMENT SPECIFICATIONS

2.3.1 Hardware requirements

1. **Processor:** Dual Core 1.60 GHz or higher
2. **Hard disk:** 500 GB
3. **RAM:** 4 GB
4. **Monitor:** 17" Colour Monitor
5. **Mouse:** Microsoft
6. **Keyboard:** Microsoft multimedia keyboard

2.3.2 Software Requirements

1. **Operating System:** Windows 10 or higher
2. **Framework:** Microsoft .Net Framework 4.5
3. **Environment:** Visual Studio 2022
4. **Front End:** C#.NET
5. **Back End:** Microsoft SQL Server 2022 or higher
6. **Documentation:** Microsoft Word 2010 or higher

2.3.3 Functional requirements

A Functional Requirement (FR) is a description of the service that the software must offer. It describes a software system or its component. A function is nothing but inputs to the software system, its behavior and outputs. It can be a calculation, data manipulation, business process, user interaction, or any other specific functionality which defines what function a system is likely to perform.

System Administrator Module

The system administrator module verifies new user registrations. Upon registration, the administrator is notified and can approve or decline the user. Once approved, the user receives an email with a system-generated password. The user can change the password after the first login

FN 1. Admin Login: Admin login manages to log into the platform and manage their profile.

Input: Username and Password.

Output: Login successful and login page is visible.

FN 2. Manage advocate: The system administrator can approve, delete and view advocate.

Input: AdvocateID, AdvocateCode, Name, Area, Date of Birth, Mobile, Email, Gender, Username, Status

Output: Approve/Reject/View advocates.

FN 3. Manage petitioner: The system administrator can approve, delete and view Petitioner.

Input: PetitionerID, PetitionerNo, Name, Date of Birth, Mobile, Email, Gender, Username, Status

Output: Approve/Reject/View Petitioners.

FN 4. Manage Court officer: The system administrator can add delete and view court officer.

Input: Username, Gender, Name, Contact No, Email ID, DOB, Password, court Type, AdvocateCode, Address, Judge ID

Output: Judge added successfully.

FN 5. Manage Police Officer: The system administrator can approve, delete and view police officer.

Input: PoliceID, Police Code, Name, Police station Date of Birth, Mobile, Email, Gender, Username, Status

Output: Approve/Reject/View Police officers.

FN 6. Mail users: can mail various users in the system and their credentials.

Input: Mail ID, Date , Username, Subject, Message, Attachment.

Output: Mail send.

Petitioner Module

The petitioner module allows the public to register, log in, and report incidents, receiving a system-generated incident ID. The petitioner can then choose a lawyer from the registered list based on their profiles and send a request with the incident ID. If the lawyer accepts and shares the e-vakalathu, the petitioner can download it and communicate further through the website. If declined, the petitioner is notified and can select a different lawyer.

FN 1. Register: It allows the petitioner to register to the platform to access its features

Input: Username, Gender, Name, Contact number, Mail id, Address

Output: petitioner registered successfully.

FN 2. petitioner Login: Petitioner login manages to log into the platform and manage their profile.

Input: Username and Password.

Output: Login successful and login page is visible.

FN 3. File Petition: Enables the petitioner to report the incident.

Input: Incident id, Username, Incident type, Police station, Petition date, Incident date, Incident area, Incident details, Attachment

Output: Incident submitted successfully

FN 4. Petition Status: Enables the petitioner to to see the petition status.

Input: Petition ID

Output: Petition status.

FN 5. Assign Advocate: Used to select a lawyer from the list of lawyers

Input: Case ID, FIR No, Case Type, Police station, Case date, FIR date, IPC Section, Petition Username, Incident Area, Respondent Username, Incident Date, Police username, User Type, FIR details, Advocate username

Output: Advocate request send.

FN 6. View case status: Petitioner can view case status.

Input: case ID.

Output: Case details are visible

FN 7. E-vakalathu: enables users to view the digitally generated e-vakalat that is being shared by the lawyer and download it if required.

Input: Petition ID

Output: E-vakalathu visible.

FN 8. Advocate History: Petitioner can view case status.

Input: Username.

Output: Advocate profile and history is visible.

FN 9. Hearing Details: can view Hearing schedules and deatils.

Input: Case ID.

Output: Hearing details are visible.

FN 10 . Mail users: can view various users in the system and their credentials.

Input: Mail ID, Date, Username, Subject, Message, Attachment.

Output: Mail send.

Police Officer

The police module represents the general police station. Upon registering into the website and then logging into the website, he/she can verify upload by the petitioner, the incident to the website after which the user will be provided with a system-generated incident ID. The police verify the complaint and generate FIR and send to the court for the further procedures.

FN 1. Register: It allows the police to register to the platform to access its features

Input: username, gender, name, contact number, email id, Police officer id, address

Output: petitioner registered successfully.

FN 2. Police officer Login: Police officer login manages to log into the platform and manage their profile.

Input: username and password.

Output: Login successful and login page is visible.

FN 3. View case status: Police can view case status.

Input: case ID.

Output: Case details are visible

FN 4. Court petition: Police officer verifies court petition.

Input: Petition id. Petition type, Username, Police station, Petition Date, Incident Date, Incident Area, Incident details

Output: Petition close/Register Case.

FN 5. File Case: Police officer files case.

Input: Petition id. Case ID, case type, Username, Police station, Petition Date, IPC Section, Petitioner username, Incident Area, Respondent Username, Incident date, Court officer, Incident details

Output: Petition close/Register Case

FN 6. Mail users: can view various users in the system and their credentials.

Input: Mail ID, Date, Username, Subject, Message, Attachment.

Output: Mail send.

FN 7. Add Evidence: Police helps to submit evidence to court.

Input: Case id, FIR No, Evidence Type, Petitioner Username, Evidence Date, Advocate Username, Evidence, Attachment.

Output: Evidence uploaded successfully.

Court Officer Module

The court officer module is responsible for scheduling the duties of the judge. He/She is a designated court official. He/she will be responsible for notifying the judge about the cases to the attended on a particular day, sending reminders to the judge for upcoming important cases as well as passing any additional information to the judge like case transfers between courts, etc.

FN 1. Court officer Login: Court officer login manages to log into the platform and manage their profile.

Input: username and password.

Output: Login successful and login page is visible.

FN 2. Approve petitions: The court officer approves petitions as submitted by the petitioner.

Input: Incident id. Petition type, Username, Police station, Petition Date, Incident Date, Incident Area, Incident details

Output: petition allocated/rejected successfully.

FN 3 Case First schedule: The court officer assigns first schedule date.

Input: Case ID, Case No, Case Type, Police station, Case date, IPC Section, Petition Username, Incident Area, Respondent Username, Incident Date, Police username, User

Type, Respondent Advocate username, Petitioner advocate username, First hearing Date, Hearing type.

Output: Petition allocated/rejected successfully

FN 4. Schedule hearing: enables to schedule hearing of various cases and can view the details of hearing as well as update them as the case forwards.

Input: Case ID, Case Type, Hearing ID, Police station, Case Status Petition Date, Hearing Date, Hearing type, Respondent Username, Hearing details, Petitioner advocate, Respondent advocate, Incident details, Hearing Date, Hearing Type

Output: Hearing scheduled successfully.

FN 5. View evidence: The evidence submitted by lawyer is directly accessible by the court officer which is submitted to court.

Input: Case ID.

Output: Evidence is visible.

FN 6. Mail users: can view various users in the system and their credentials.

Input: Mail ID, Date, Username, Subject, Message, Attachment.

Output: Mail send.

Advocate Module

The advocate module allows lawyers to accept or decline petitioner requests to represent cases. Upon acceptance, the lawyer can view incident details using the provided incident ID, generate an e-vakalat, and send it to the petitioner. Lawyers can also communicate with petitioners and other lawyers, and access previous case details from the website's database.

FN 1. Advocate Login: Lawyer login manages to log into the platform and manage their profile.

Input: username and password.

Output: Login successful and login page is visible.

FN 2. Approve Case: Enables the lawyers to accept or decline the request put forward by petitioners to attend their cases.

Input: Case ID, FIR No, Case Type, Police Station, Case Date, FIR Date, IPC Section, Petition Username, Incident Area, Respondent Username, Incident Date, Police Username

Output: Accept or decline case request.

FN 3. Search Incident: Enables the lawyer to access the details of an incident reported by the petitioner using the incident ID shared by the petitioner.

Input: Petition id.

Output: Incident details are visible.

FN 4. Add Evidence: Lawyer helps to submit evidence to court.

Input: Case id, FIR No, Evidence Type, Petitioner Username, Evidence Date, Advocate Username, Evidence, Attachment.

Output: Evidence uploaded successfully.

FN 4. Status of case: Used to view status of the case.

Input: Case id.

Output: Case status is visible.

FN 5. Hearing details: Lawyer can view the details of case hearing as declared by the judge through court admin.

Input: case id.

Output: Hearing details is visible.

FN 6. Mail users: Can view various users in the system and their credentials.

Input: Mail ID, Date, Username, Subject, Message, Attachment.

Output: Mail send.

2.3.4 Non-Functional Requirements

In systems, a non-functional requirement (NFR) is a requirement that specifies criteria that can be used to judge the operation of a system, rather than specific behaviour. They are contrasted with functional requirements that define specific behaviour or functions:

- **Usability:** Usability defines how difficult it will be for a user to learn and operate the system. Efficiency of use: the average time it takes to accomplish a user's goals, how many tasks a user can complete without any help, the number of transactions completed without errors
- **Authenticity:** The system meets authenticity as only those users with a valid username and password can log into the application.
- **Data integrity:** Data integrity is ensured by only the accurate data.
- **Performance:** It ensures good performance by maintaining a low response time, high utilization and fast throughput.
- **Availability:** The application meets availability as a user can access it any time.
- **Security:** The data is securely stored in database and only the authorized user can access it

2.4 FEASIBILITY STUDY

Feasibility study is the test of the system proposal made to identify whether the user needs may be satisfied using the current software and hardware technologies, whether the system will be cost effective from a business point of view and whether it can be developed with the given budgetary constraints. A feasibility study should be relatively cheap and done at the earliest possible time. Depending on the study, the decision is made whether to go ahead with a more detailed analysis. When a new project is proposed, it normally goes through feasibility assessment. Feasibility study is carried out to determine whether the proposed system is possible to develop with available resources and what should be the cost consideration. Facts considered in the feasibility analysis were:

- Economic Feasibility
- Operational Feasibility
- Technical Feasibility

2.4.3 Economic Feasibility

This feasibility study present tangible and intangible benefits from the prefect by comparing the development and operational cost. This system needs some more initial investment than the existing system, but it can be justifiable that it will improve quality of service. Thus, feasibility study should centre along the following points:

- Improvement resulting over the existing method in terms of accuracy, timeliness.
- Cost comparison
- Estimate on the life expectancy of the hardware
- Overall objective Our project is economically feasible.

It does not require much cost to be involved in the overall process. It is essential because the main goal of the proposed system is to have economically better result with the increase.

2.4.4 Operational Feasibility

Operational analysis is the most frequently used method for evaluating the effectiveness of a new system. More commonly known as cost/benefit analysis, the procedure is to determine the benefits and saving that are expected from a candidate system and compare them with costs. If benefits outweigh costs, then the decision is made to design and implement the system. An entrepreneur must accurately weigh the cost versus benefits before taking an action. Cost-based study: It is important to identify cost and benefit factors, which can be categorized as follows:

1. Development costs.
2. Operating costs.

This is an analysis of the costs to be incurred in the system and benefits derivable out of the system. Time-based study: This is an analysis of the time required to achieve a return on investments The future value of a project is also a factor. This application can be operated by all the users across the world.

2.4.5 Technical Feasibility

Technical Feasibility deals with the hardware as well as software requirements. Technology is not a constraint to type system development. We have to find out whether the necessary technology, the proposed equipment has the capacity to hold the data, which is used in the project, should be checked to carry out this technical feasibility. The technical feasibility issues usually raised during the feasibility stage of investigation includes

- This software is running in Windows 11.
- The minimum hardware required is 3GB and maximum hardware required is 400GB.
- The system can be expanded

SYSTEM DESIGN

3.1 DESIGN OF SUBSYSTEM

A Design of Subsystem encapsulates behaviour, providing explicit and formal interfaces, and does not expose its internal contents. This provides the ability to completely encapsulate the interactions of a number of classes and/or subsystems. This project has five major modules. These five modules are the core functionalities of the system. The project is divided into five major modules according to their functionalities.

1. ADMIN
2. PETITIONER
3. COURT OFFICER
4. ADVOCATE
5. POLICE OFFICER

1. Admin Module:

The system admin can login and manage various users in the system. The admin is responsible for approving/deleting users based on their profile. Admin is also responsible for adding court officer into the system. The admin can view various users and individual members if required

- **Admin Login:** Admin login manages to log into the platform and manage their profile.
- **Manage Advocates:** Admin can approve, delete and view advocates.
- **Manage Petitioners:** Admin can approve, delete and view petitioners.
- **Manage court officers:** Admin can add, delete and view court officers.
- **Manage Police officers:** Admin can approve, delete and view police officers.
- **Mail:** Admin can view and compose mails to various users.

2. Petitioner Module:

The Petitioner module is the user module that enables to access judiciary services regarding his/her case. It enables to submit petition and watch update without going through various offline restrictions. The petitioner can also request an advocate through the system. Petitioner can mail a different user if necessary.

- **Registration:** Petitioner can create account.
- **Login:** The petitioner can log on to the system using valid login credentials and make use of the functionalities provided by the website.

- **Report incident:** Enables the petitioner to report the incident and after doing so will receive a system generated petition ID.
- **View case status:** Petitioner can view his/her case status.
- **Select advocate:** Used to select a lawyer from the list of lawyers after viewing each of their profile in order to represent their case in court.
- **Mail:** used to communicate with the other users of the website.
- **View e-vakalathu:** enables users to view the digitally generated e-vakalathu that is being shared by the lawyer and download it if required

3. Court officer module

Court officer is the only user module which cannot register itself rather added by the system admin. Court officer represents the corresponding judge he/she is working for. Court officer can update case details regarding hearing as per the judgement forwarded by the judge. He/she can also see evidence details through the system without going through hectic methods.

- **Login:** The admin should login to the system using valid login credentials to monitor and carry out various functionalities.
- **Approve petitions:** The court officer approves petitions as submitted by the petitioner.
- **Hearing scheduling:** enables to schedule hearing of various cases and can view the details of hearing as well as update them as the case forwards.
- **View evidence:** The evidence submitted by lawyer is directly accessible by the court officer which is submitted to court.
- **Mail:** Enables court officer to communicate with various user as required.

4. Advocate module

Advocate module enables advocates to approve requests of various users. Advocates can manage their cases by easily accessing every case details and the system ensures that maximum transparency. Advocate can easily upload evidences with just one click. Also enables them to summarise case status without going through multiple enquiries

- **Login:** The lawyer can log on to the system using valid login credentials and make use of the functionalities provided by the website.
- **Accept/decline client requests:** Enables the lawyers to accept or decline the request put forward by petitioners to attend their cases.

- **Search Incident:** Enables the lawyer to access the details of an incident reported by the petitioner using the incident ID shared by the petitioner.
- **Add Evidence:** Lawyer helps to submit evidence to court.
- **Status of case:** used to view status of the case.
- **Hearing details:** Lawyer can view the details of case hearing as declared by the judge through court admin.
- **Mail:** enables the user to communicate with other users of the website.

5. Police officer module

Police officer module allows police to approve court petitions and file them without going through hard processes of manually collecting case details. The police can know and verify the status of the existing active cases easily. It ensures maximum transparency and avoids cases of non-recorded case settlements.

- **Registration:** The police register information such as the username, station, etc and also update his/her profile
- **Login:** The police can log on to the system using valid login credentials and make use of the functionalities provided by the website Case management
- **View case status:** The police can view status of various cases by their case id
- **Verify petitions:** The police verify the petition and petitioner.
- **Mail :** used to communicate with the other users of the website.

3.2 USER INTERFACE DESIGN

3.2.1 Input Design of User

Input designing is the basic theory to be considered during system study. The input media used in the system is the keyboard. Details are entered in the system through different data entry screens. The system is designed in a user-friendly manner. Appropriate error messages are displayed when a false data is entered. The user interface design is very important for any application. The interface design defines how the software communicates within itself, to system that interpreted with it and with human who use it. The input design is the process of converting the user-oriented description of inputs into a programmer-oriented specification. It is the link that ties the system into the world of its users. Input design involves determining the record media, method of input, speed of capture and entry to the system. The analyst should consider the following points when designing the input:

- Nature of the input processing.
- Flexibility and thoroughness validation rules.
- Handling of priorities with the input procedures.
- Use of composite input documents to reduce the number of different ones.

Login page

Public can view case status?
requires a free account and provides the same access as Public View.

[VIEW CASES](#)

1	SUPREME COURT	25	HIGH COURTS
672	DISTRICT COURTS	14	TRIBUNALS

The proceedings of the Supreme Court are conducted in English only. **Quality Service**
Supreme Court Rules, 1966 are framed under Article 145 of the Constitution to regulate the practice and procedure of the Supreme Court.
Call Now For Immediate Assistance

Username:
Password:

[Login](#) [Cancel](#)

3.2.2 Output Design of User

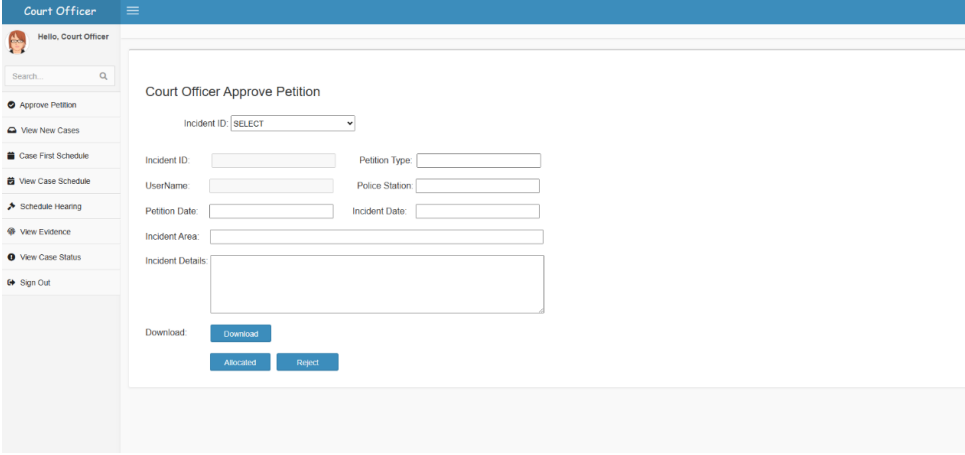
An inevitable activity in the system design is the proper design of output in a form acceptable to the user. Outputs from the system are required primarily to communicate the result of processing to users. Outputs also provide a permanent copy of the results for later consultation. An intelligible output will improve system relationship with the user and help in the decision making process.

The approach to output design is very dependent on the type of output and nature of data. Special attention has to be made to data editing. The choice of appropriate output medium is also an important task. The selection may be affected by the following kinds of consideration:

- Response time.
- Location of users.
- Cost.
- Software/hardware.
- Suitability of the device for application concerned.

The output design must be specified and documented, data items have to be accurately defined and arranged for clarity and easy comprehension.

Court officer interface



The screenshot displays a web application interface for a Court Officer. The top navigation bar is blue with the text 'Court Officer' and a hamburger menu icon. Below this, a sidebar on the left contains a search bar and a list of menu items: 'Approve Petition' (selected), 'View New Cases', 'Case First Schedule', 'View Case Schedule', 'Schedule Hearing', 'View Evidence', 'View Case Status', and 'Sign Out'. The main content area is titled 'Court Officer Approve Petition'. It features a form with the following fields: 'Incident ID' (a dropdown menu with 'SELECT' as the current value), 'Incident ID' (a text input field), 'Petition Type' (a text input field), 'UserName' (a text input field), 'Police Station' (a text input field), 'Petition Date' (a text input field), 'Incident Date' (a text input field), 'Incident Area' (a text input field), and 'Incident Details' (a large text area). At the bottom of the form, there is a 'Download' section with three buttons: 'Download', 'Allocated', and 'Reject'.

3.3 DATABASE DESIGN

The overall objective in the development of database technology has been to treat data as an organizational resource and as an integrated whole. DBMS allow data to be protected and organized separately from other resources. Database is an integrated collection of data. The most significant form of data as seen by the programmers is data as stored on the direct access storage devices. This is the difference between logical and physical data.

Database files are the key source of information into the system. It is the process of designing database files, which are the key source of information to the system. The files should be properly designed and planned for collection, accumulation, editing and retrieving the required information. The organization of data in database aims to achieve three major objectives:

- Data integration.
- Data integrity.
- Data independence.

The suggested system keeps the data employed in processing in the MySQL database. MySQL is said to be an open-source relational database management system (RDBMS) conforming to the Structured Query Language (SQL) standard. As opposed to NoSQL databases, MySQL is a traditional relational database where data is kept in tables of defined schemas in an organized manner. MySQL uses a master-slave replication model to achieve data consistency and availability. Under this model, the master database does all the write operations, and the slave databases maintain copies of the master database. Changes to the master database are replicated automatically to the slave databases to provide redundancy and allow for automatic failover in case of a failure. In MySQL, data is stored in tables, which consist of rows and columns. A column represents a specific field, while a row represents a single record. Tables are related to each other by the use of primary and foreign keys, thus maintaining data integrity and reducing redundancy. Every row in a MySQL table has a primary key, which is used to uniquely identify every row. MySQL has data types such as integers, floating points, strings, date/time, and binary. MySQL also supports transactions, stored procedures, and triggers for enforcing data consistency and business logic.

To develop applications, ASP.NET can be utilized to communicate with MySQL databases and develop dynamic web applications. ASP.NET, a Microsoft-developed framework, enables

developers to develop web applications and APIs in C#. Database connectivity, authentication, and security are natively supported by ASP.NET. MySQL can be combined with ASP.NET with the help of Entity Framework (EF), an Object-Relational Mapper (ORM), which enables developers to utilize objects in C# to communicate with databases rather than executing raw SQL queries. ASP.NET applications can perform database operations in ASP.NET Core and MySQL, which supports CRUD operations. ASP.NET MVC or ASP.NET Web API may be utilized by developers to develop web-based applications to interact with MySQL databases efficiently.

MySQL databases are subject to querying through SQL statements, including SELECT, INSERT, UPDATE, and DELETE, which facilitate efficient data manipulation. Furthermore, intricate queries may be conducted through the implementation of JOIN operations, stored procedures, and indexing, thereby enhancing performance optimization. Utilizing MySQL as the database and ASP.NET for application development, the system offers robust, structured, and expandable data management for web applications.

Normalization

Normalization is a method of finding redundant fields and splitting a big table into smaller tables that are manageable. It also serves the purpose of minimizing insertion, deletion, and update anomalies. This table is normalized to the third normal form. Normalization is a procedure of organizing a relational database based on a specific sequence of normal forms with the goal of minimizing data redundancy and enhancing data integrity. Normalization is the process of aligning columns (attributes) and tables (relations) of a database in a way that their dependencies are properly maintained by the integrity constraints of the database. Normalization is achieved through the application of pre-established rules, and this can be accomplished through synthesis (designing a new database schema) or decomposition (enhancing an existing database schema). The rules that are used to direct normalization are key to making the database free from redundancy, inconsistencies, and other issues. Essentially, these rules offer the avenue to split huge, complicated tables into manageable pieces without sacrificing the data dependencies.

The database architecture has been designed to adhere to the following criteria:

- Data is organized in a manner that avoids redundancy. Breaking a big table into small tables avoids repetition and data entry, deletion and updation anomalies.
- We ensure that all the non-key attributes are functionally dependent on the primary key. This avoids partial dependency complexities, where attributes are functionally dependent on a subset of the primary key, especially when the primary key is composite, made up of more than one attribute.
- The design avoids transitive dependencies. That is, non-key columns cannot be dependent on other non-key columns, which forms indirect data relations that are hard to control and update. Functional dependencies are respected, so that the values of a given attribute are always derived from other attributes, leading to better integrity and correctness of the data that is stored in the database.

In short, the rules for each of the five normal forms are as follows: -

- First normal form

A relation is said to be in 1NF if all the underlying domain of attributes contain simple individual values.

- Second normal form

The 2NF is based on the concept of full functional dependency. A relation said to be in 2NF if and only if it is in 1NF and every non-key attribute is fully functionally dependent on candidate key of the table.

- Third normal form

The 3NF is based on the concept of transitive dependency. A relation in 2NF is said to be in 3NF if every non-key attribute is non-transitively dependent on candidate key of the table. Our project is in the second normal form because every non key attribute is fully functionally dependent on primary key. A relation is in third normal form, if there is no transitive dependency for non-prime attributes as well as it is in second normal form.

- Boyce Codd normal form (BCNF)

BCNF is the advance version of 3NF. It is stricter than 3NF. A table is in BCNF if every functional dependency $X \rightarrow Y$, X is the super key of the table. For BCNF, the table should be in 3NF, and for every FD, LHS is super key.

- Fourth normal form (4NF)

A relation will be in 4NF if it is in Boyce Codd normal form and has no multi-valued dependency. For a dependency $A \twoheadrightarrow B$, if for a single value of A, multiple values of B exist, then the relation will be a multi-valued dependency.

- Fifth normal form (5NF)

A relation is in 5NF if it is in 4NF and not contains any join dependency and joining should be lossless. 5NF is satisfied when all the tables are broken into as many tables as possible in order to avoid redundancy. 5NF is also known as Project-join normal form (PJ/NF).

Our database us in 2NF.

3.3.1 Table Design

Advocate

Field Name	Datatype	Constraint
Adid	Int	PRIMARY KEY
Adcode	Varchar(50)	NOT NULL
Name	Varchar(50)	NOT NULL
Adtype	Varchar(50)	NOT NULL
Dob	Varchar(50)	NOT NULL
Address	Varchar(MAX)	NOT NULL
Phone	Varchar(50)	NOT NULL
Email	Varchar(50)	NOT NULL
Gender	Varchar(50)	NOT NULL
Username	Varchar(50)	NOT NULL
Pwd	Varchar(50)	NOT NULL
Certificate	Varchar(MAX)	NOT NULL
Status	Int	NOT NULL
Aphoto	Varchar(MAX)	NOT NULL

Maildetails

Field Name	Datatype	Constraint
Mid	Int	PRIMARY KEY
mdate	Varchar(50)	NOT NULL
mform	Varchar(50)	NOT NULL
mtto	Varchar(50)	NOT NULL
subject	Varchar(50)	NOT NULL
attachment	Varchar(MAX)	NOT NULL
message	Varchar(MAX)	NOT NULL
status	Int	NOT NULL

Case details

Field Name	Datatype	Constraint
caseid	Int	PRIMARY KEY
casetype	Varchar(50)	NOT NULL
ipcsection	Varchar(50)	NOT NULL
firno	Varchar(50)	NOT NULL
casedate	Varchar(50)	NOT NULL
firdate	Varchar(50)	NOT NULL
Fir	Varchar(MAX)	NOT NULL
incidate	Varchar(50)	NOT NULL
inilocation	Varchar(50)	NOT NULL
puname	Varchar(50)	NOT NULL
petuname	Varchar(50)	NOT NULL
respuname	Varchar(50)	NOT NULL
petadvuname	Varchar(50)	NOT NULL
resadvuname	Varchar(50)	NOT NULL
uploaddetails	Varchar(50)	NOT NULL
status	Int	NOT NULL
petid	Int	NOT NULL

Login

Field Name	Datatype	Constraint
lid	int	PRIMARY KEY
username	Varchar(50)	NOT NULL
pwd	Varchar(50)	NOT NULL
status	int	NOT NULL

Evidence

Field Name	Datatype	Constraint
evid	Int	PRIMARY KEY
caseid	Int	FOREIGN KEY
firno	Varchar(50)	FOREIGN KEY
petuname	Varchar(50)	NOT NULL
petadvuname	Varchar(50)	NOT NULL
evtype	Varchar(50)	NOT NULL
evdetails	Varchar(MAX)	NOT NULL
evupload	Varchar(MAX)	NOT NULL
evdate	Varchar(50)	NOT NULL
status	Int	NOT NULL

Hearing

Field Name	Datatype	Constraint
Hid	Int	PRIMARY KEY
firno	Int	FOREIGN KEY
petuname	Varchar(50)	NOT NULL
petadvuname	Varchar(50)	NOT NULL
respuname	Varchar(50)	NOT NULL
respadvuname	Varchar(50)	NOT NULL
heardate	Varchar(MAX)	NOT NULL
heartype	Varchar(MAX)	NOT NULL
details	Varchar(50)	NOT NULL
status	Int	NOT NULL
cstatus	Varchar(50)	NOT NULL

Judge

Field Name	Datatype	Constraint
jd	int	PRIMARY KEY
jcode	Varchar(50)	NOT NULL
name	Varchar(50)	NOT NULL
ctype	Varchar(50)	NOT NULL
dob	Varchar(50)	NOT NULL
address	Varchar(MAX)	NOT NULL
phone	Varchar(50)	NOT NULL
email	Varchar(50)	NOT NULL
gender	Varchar(50)	NOT NULL
username	Varchar(50)	NOT NULL
pwd	Varchar(50)	NOT NULL
location	Varchar(50)	NOT NULL
status	int	NOT NULL

Petition

Field Name	Datatype	Constraint
petitionid	int	PRIMARY KEY
puname	Varchar(50)	NOT NULL
pdate	Varchar(50)	NOT NULL
ptype	Varchar(50)	NOT NULL
pdetails	Varchar(MAX)	NOT NULL
indate	Varchar(50)	NOT NULL
inlocation	Varchar(50)	NOT NULL
pstation	Varchar(50)	NOT NULL
documents	Varchar(MAX)	NOT NULL
status	int	NOT NULL
aduname	Varchar(50)	NOT NULL

Petitioner

Field Name	Datatype	Constraint
petid	Int	PRIMARY KEY
puname	Varchar(50)	NOT NULL
pdate	Varchar(50)	NOT NULL
pdetails	Varchar(50)	NOT NULL
indate	Varchar(50)	NOT NULL
inlocation	Varchar(50)	NOT NULL
pstation	Varchar(50)	NOT NULL
documents	Varchar(MAX)	NOT NULL
status	Int	NOT NULL
aduname	Varchar(50)	NOT NULL

Police

Field Name	Datatype	Constraint
pid	Int	PRIMARY KEY
pcode	Varchar(50)	NOT NULL
name	Varchar(50)	NOT NULL
dob	Varchar(50)	NOT NULL
address	Varchar(50)	NOT NULL
phone	Varchar(50)	NOT NULL
email	Varchar(50)	NOT NULL
gender	Varchar(50)	NOT NULL
username	Varchar(50)	NOT NULL
pwd	Varchar(50)	NOT NULL
certificate	Varchar(MAX)	NOT NULL
status	Int	NOT NULL

3.3.2 Data Dictionary

Data Dictionary is the major component in the structured analysis model of the system. It lists all the data items appearing in DFD. A data dictionary is a file or a set of files that includes a database's metadata (hold records about other objects in the database), like data ownership, relationships of the data to another object, and some other data.

Sl No	Field Name	Data Type	Constraints	Description
1	adid	Int	Primary Key	Id of advocate
2	adcode	Varchar(50)	Not Null	Advocate code
3	name	Varchar(50)	Not Null	Name of the advocate
4	adtype	Varchar(50)	Not Null	Case type of advocate
5	dob	Varchar(50)	Not Null	Date of birth of advocate
6	address	Varchar(50)	Not Null	Address of advocate
7	phone	Varchar(50)	Not Null	Phone number of advocate
8	email	Varchar(50)	Not Null	Email of advocate
9	gender	Varchar(50)	Not Null	Gender of advocate
10	username	Varchar(50)	Not Null	Username of advocate
11	pwd	Varchar(50)	Not Null	Password of advocate
12	certificate	Varchar(50)	Not Null	License certificate
13	status	Varchar(50)	Not Null	Status of advocate approval

14	aphoto	Varchar(50)	Not Null	Photo of advocate
15	caseid	Int	Primary Key	ID of case
16	casetype	Varchar(50)	Not Null	Type of case
17	ipcsection	Varchar(MAX)	Not Null	IPC section assigned
18	firno	Varchar(50)	Not Null	FIR number
19	casedate	Varchar(50)	Not Null	Date of case
20	firdate	Varchar(50)	Not Null	Date of FIR
21	fir	Varchar(50)	Not Null	FIR details
22	incidate	Varchar(50)	Not Null	Date of incidence
23	inilocation	Varchar(50)	Not Null	Location of incidence
24	puname	Varchar(50)	Not Null	Username of police username
25	petuname	Varchar(50)	Not Null	Username of petitioner
26	respuname	Int	Not Null	Respondent username
27	petadvuname	Int	Not Null	Name of petitioner's advocate
28	resadvuname	Varchar(50)	Not Null	Name of respondent's advocate
29	uploaddetails	Varchar(50)	Not Null	Details of upload
30	status	Varchar(MAX)	Not Null	Status of case
31	petid	Varchar(50)	Not Null	Id of petitioner

32	evid	Int	Primary Key	ID of evidence
33	evtype	Varchar(50)	Not Null	Type of evidence
34	evdetails	Varchar(Max)	Not Null	Details of evidence
35	evupload	Varchar(Max)	Not Null	Uploaded evidence
34	evdate	Varchar(50)	Not Null	Date of evidence
35	Status	Int	Not Null	Status of evidence
36	hid	Int	Primary Key	ID of hearing
37	heardate	Varchar(Max)	Not Null	Date of hearing
38	heartype	Varchar(Max)	Not Null	Type of hearing
39	details	Varchar(50)	Not Null	Details of hearing
40	status	Int	Not Null	Status of hearing
41	cstatus	Varchar(50)	Not Null	Whether case is active/closed
42	jid	Int	Primary Key	ID of court officer
43	jcode	Varchar(50)	Not Null	Court officer code
44	name	Varchar(50)	Not Null	Name of court officer
45	ctype	Varchar(50)	Not Null	Court type
46	dob	Varchar(50)	Not Null	Date of birth of court officer
47	address	Varchar(50)	Not Null	Address of court officer

48	phone	Varchar(50)	Not Null	Phone number of court officer
49	lid	Int	Primary Key	Login ID
50	username	Varchar(50)	Not Null	Username of various users
51	pwd	Varchar(50)	Not Null	Password of various users
52	status	Int	Not Null	Status of login
53	mid	Int	Primary Key	Mail ID
54	madate	Varchar(50)	Not Null	Date of mail
55	mfrom	Varchar(50)	Not Null	From address of mail
56	mto	Varchar(50)	Not Null	To address of mail
57	subject	Varchar(50)	Not Null	Subject of mail
58	attachment	Varchar(Max)	Not Null	File attachments in mail
59	message	Varchar(Max)	Not Null	Message in mail
59	status	Int	Not Null	Status of mail
60	petitionid	Int	Primary Key	ID of petition
61	pdate	Varchar(50)	Not Null	Date of petition
62	pptype	Varchar(50)	Not Null	Type of petition
63	pdetails	Varchar(50)	Not Null	Details of petition
64	indate	Varchar(50)	Not Null	Date of incident

65	inlocation	Varchar(50)	Not Null	Location of incident
66	pstation	Varchar(50)	Not Null	Police station
67	documents	Varchar(50)	Not Null	Documents of petition
68	status	Int	Not Null	Status of petition
69	aduname	Varchar(50)	Not Null	Username of advocate
70	pid	Int	Primary Key	ID if police
71	pcode	Varchar(50)	Not Null	Police code
72	name	Varchar(50)	Not Null	Name
73	dob	Varchar(50)	Not Null	Date of birth of police officer
74	address	Varchar(50)	Not Null	Address of police officer
75	phone	Varchar(50)	Not Null	Phone number of police station
76	email	Varchar(50)	Not Null	Email of Police officer
77	gender	Varchar(50)	Not Null	Gender of Police officer
78	username	Varchar(50)	Not Null	Username of police officer
79	pwd	Varchar(50)	Not Null	Password of login
80	certificate	Varchar(Max)	Not Null	License certificate

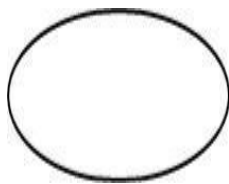
3.4 MODELLING

3.4.1 Data flow diagram (DFD)

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modelling its process aspects. A DFD is often used as a preliminary step to create an overview of the system, which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design). A DFD shows what kind of information will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of process or information about whether processes will operate in sequence or in parallel.

Data Flow Diagram Notations

Function:



File/Database:



Input/output



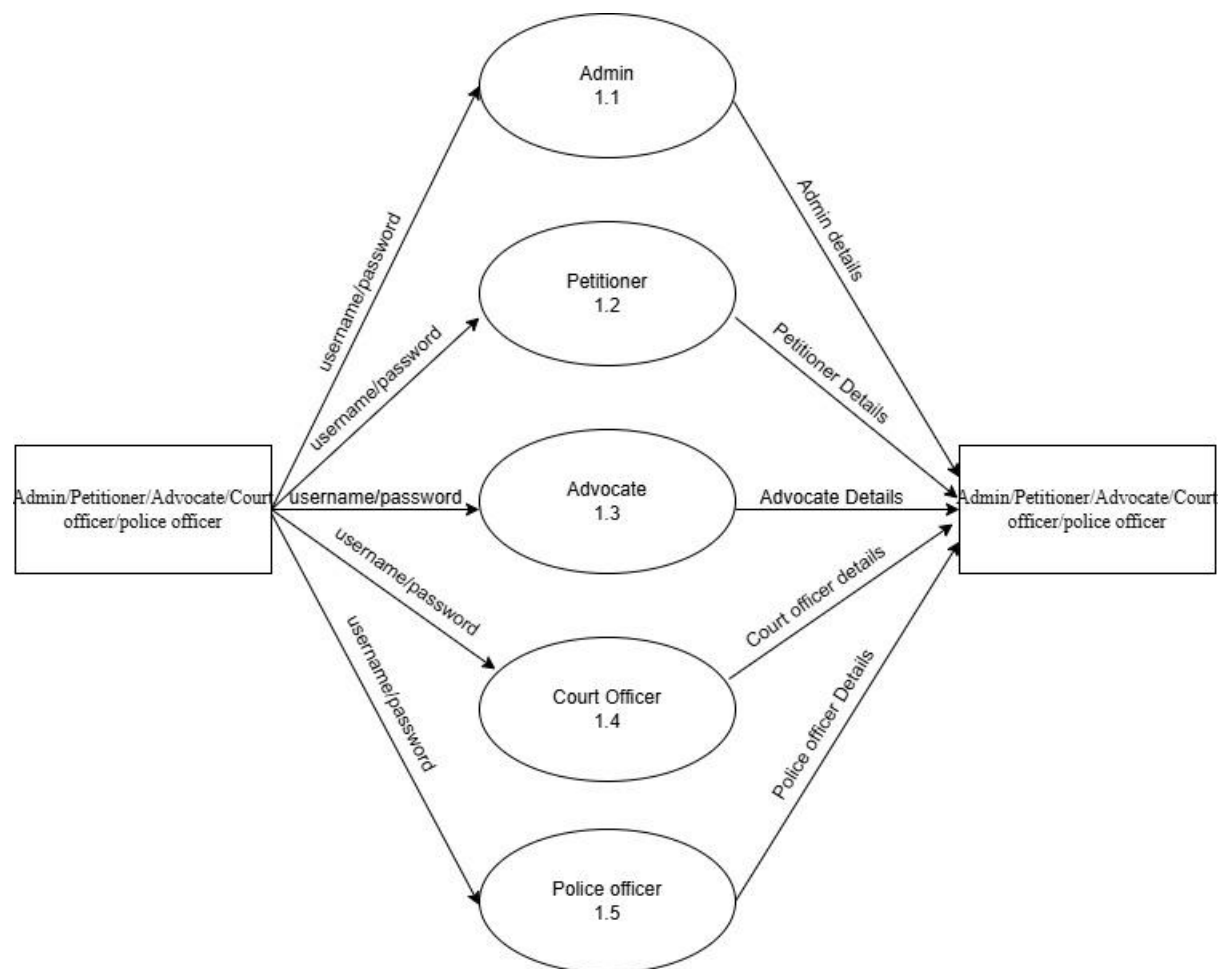
Flow of direction



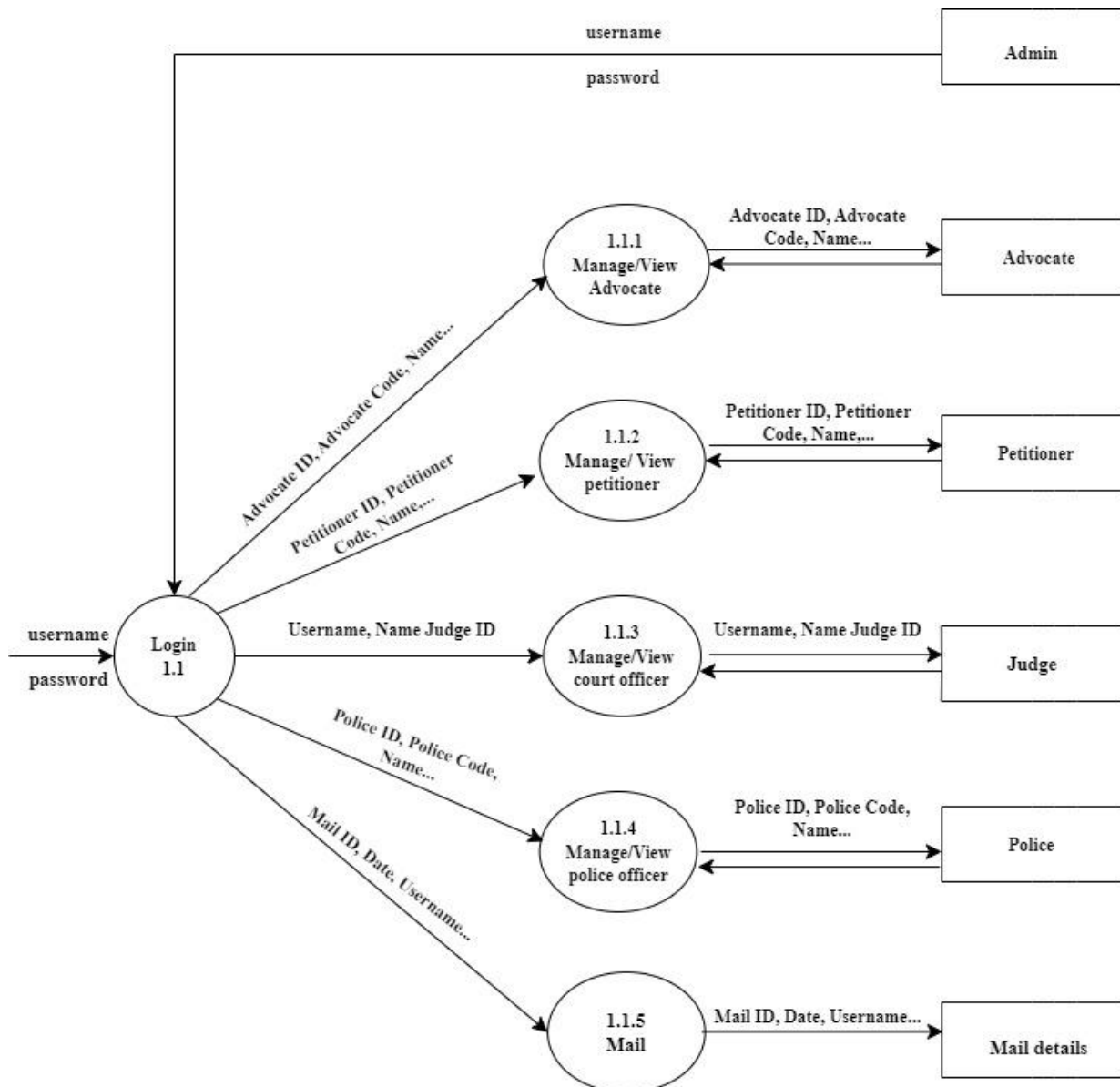
DATA FLOW DIAGRAM

LEVEL-0/CONTEXT LEVEL

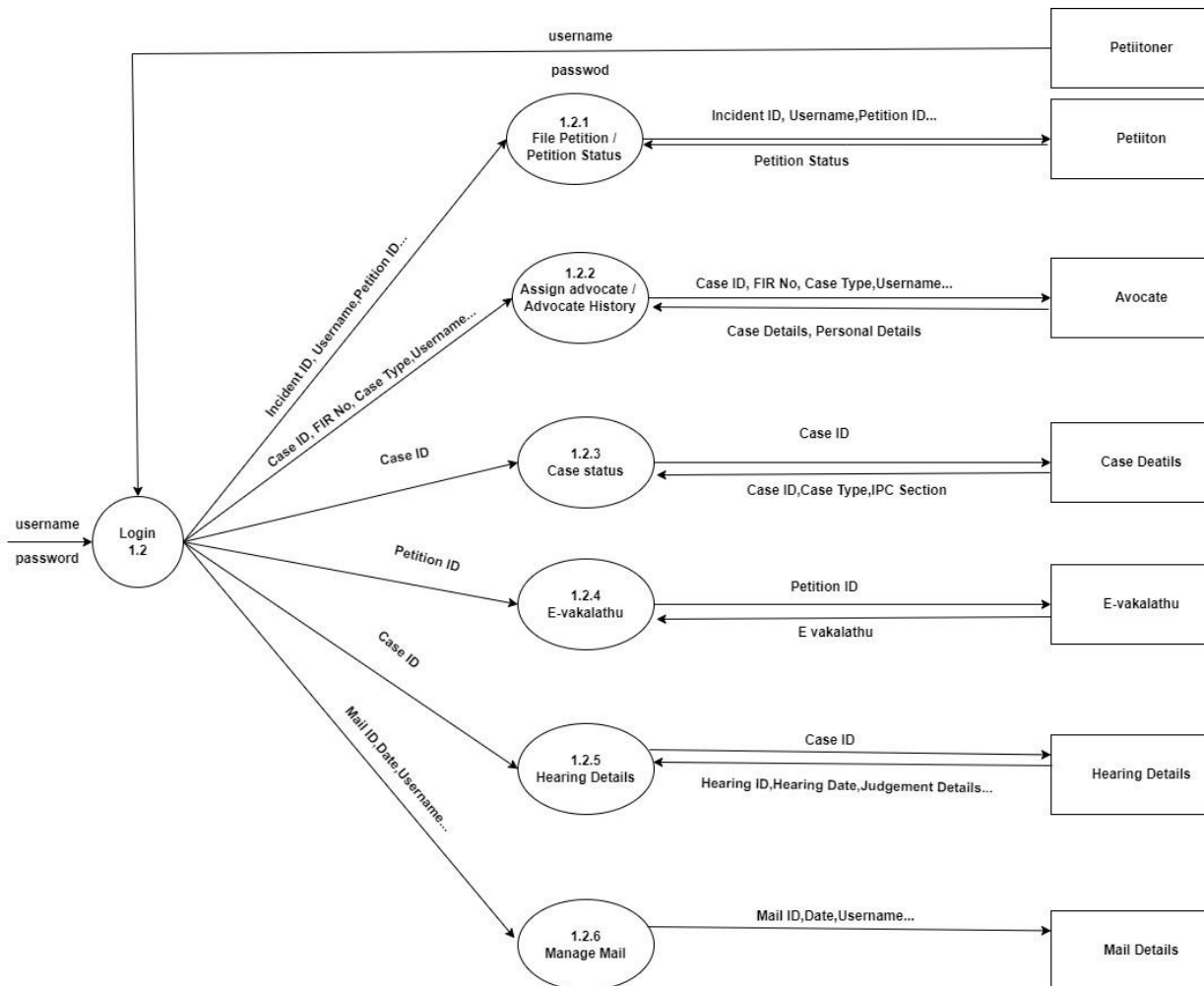


LEVEL-1

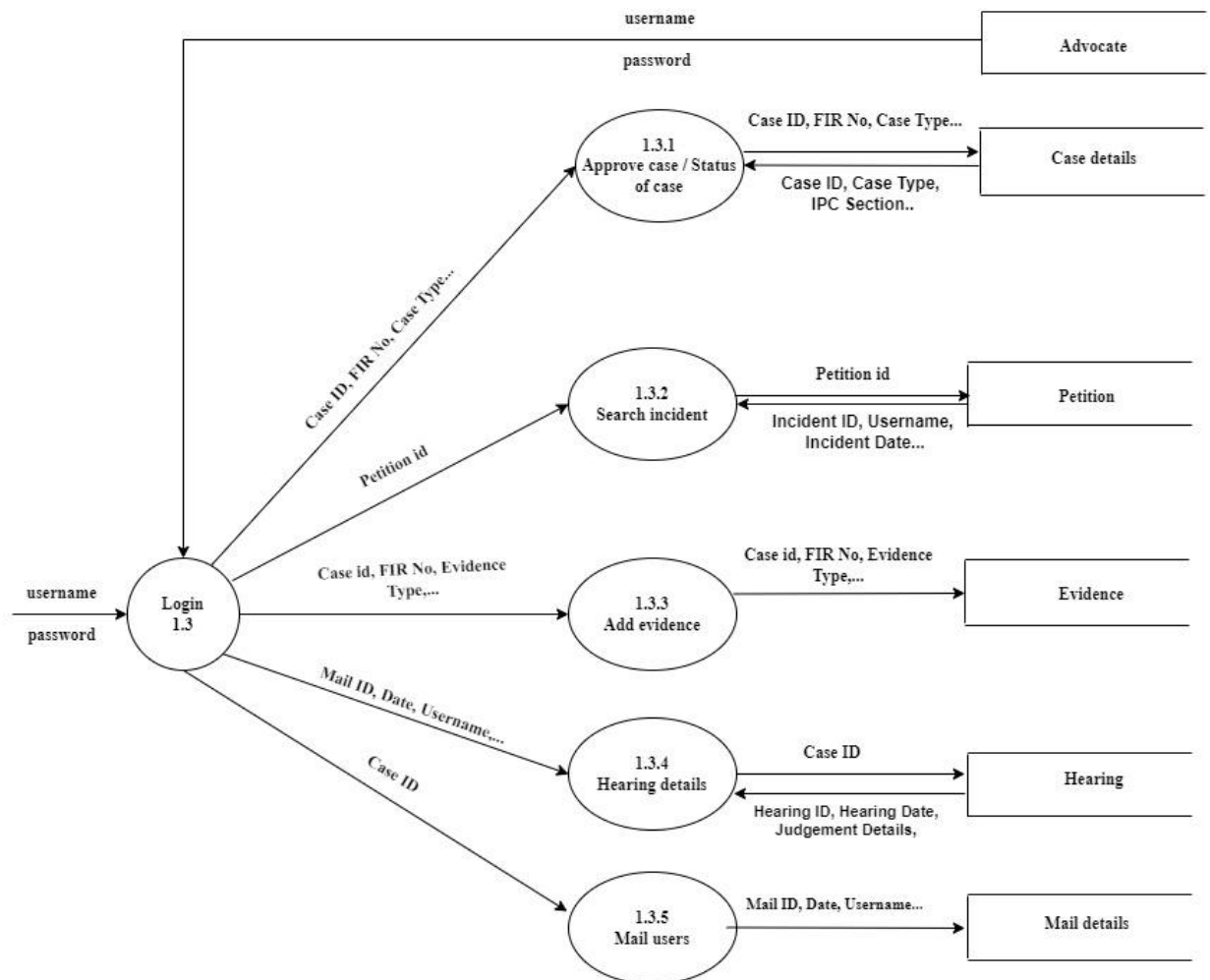
LEVEL -1.1 OF ADMIN



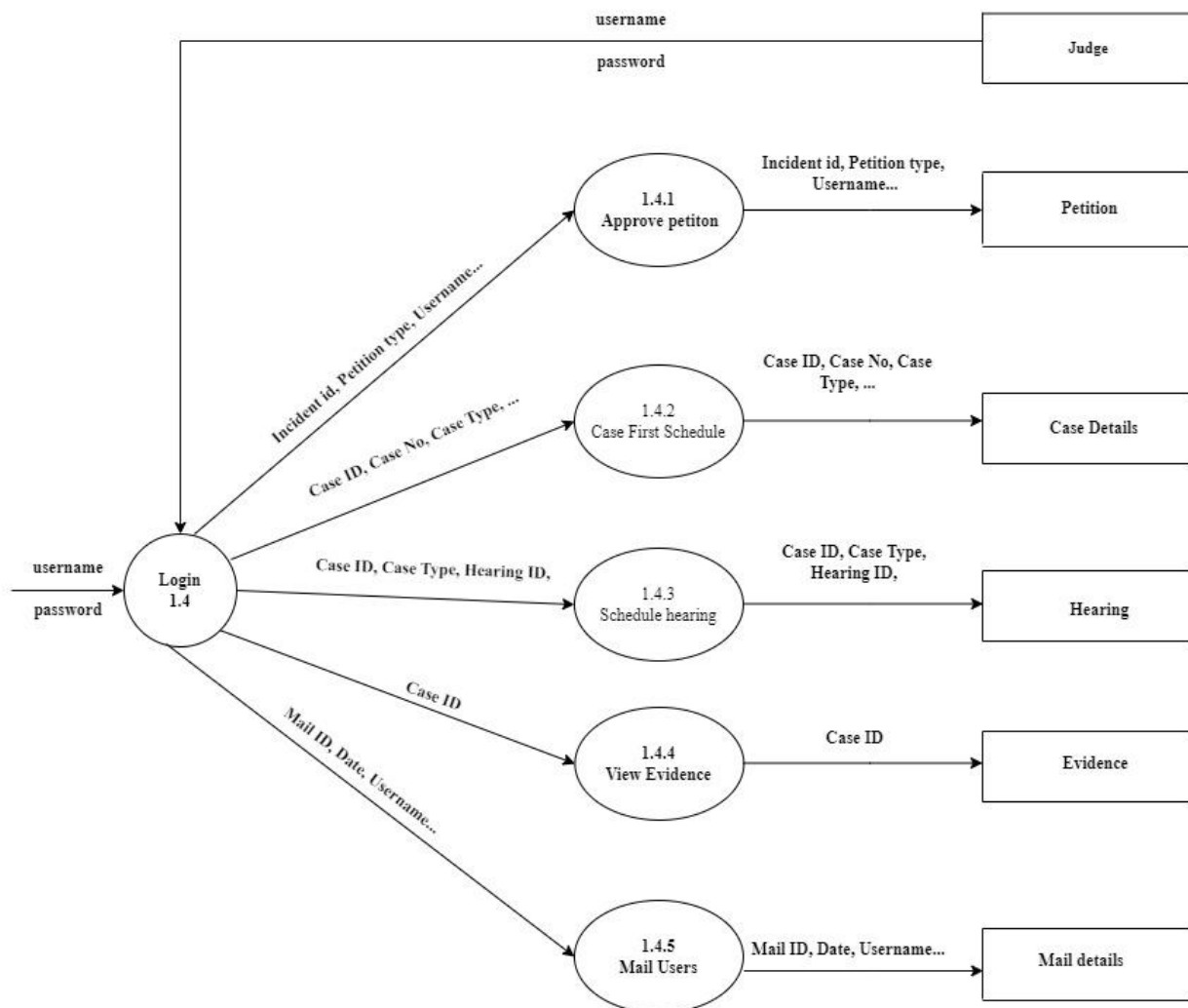
LEVEL -1.2 OF PETITIONER



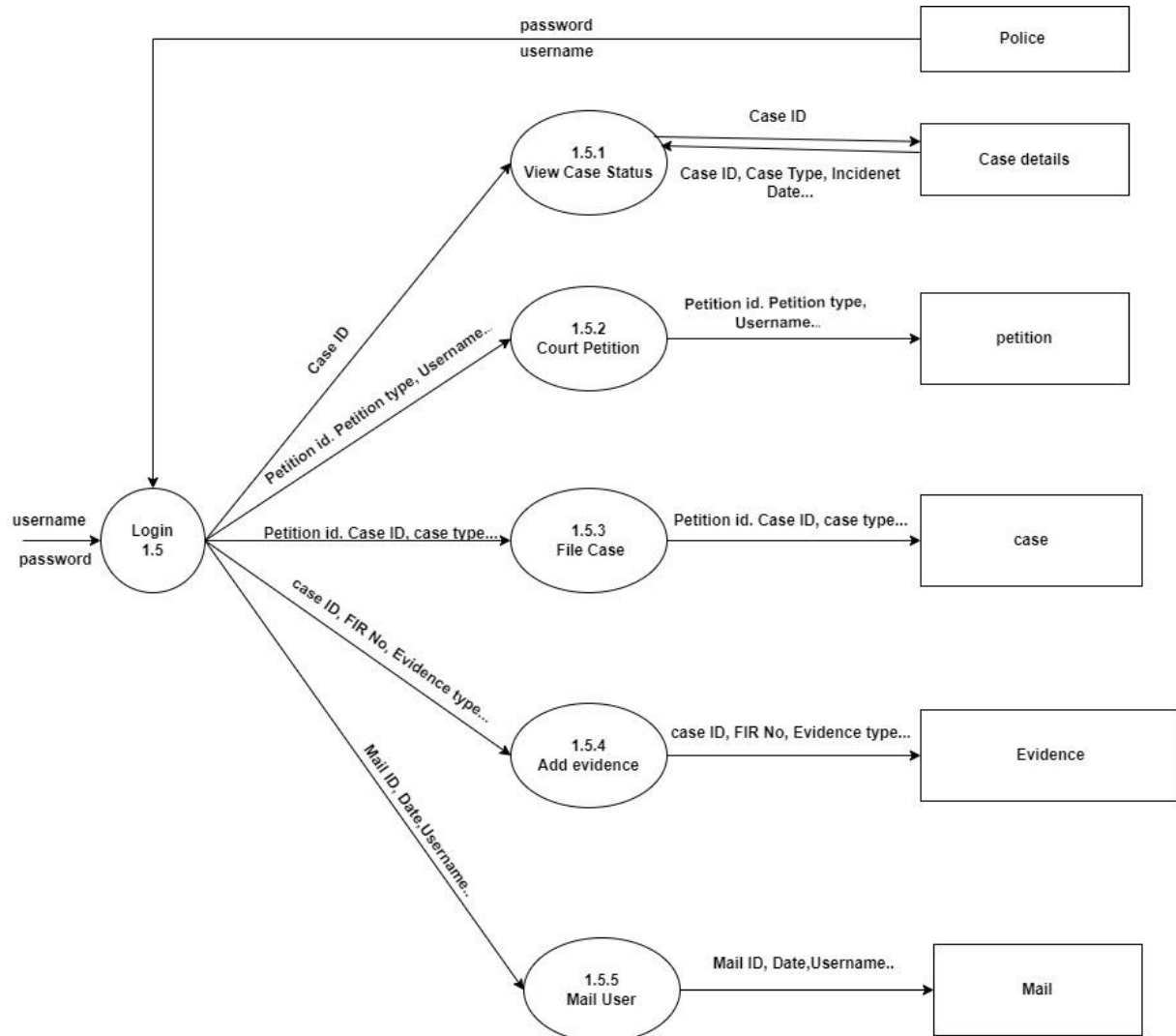
LEVEL-1.3 OF ADVOCATE



LEVEL -1.4 OF COURT OFFICER



LEVEL 1.5 OF POLICE OFFICER



3.4.2 Entity relationship model

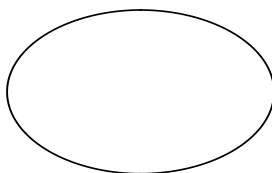
An entity relationship diagram is a specialized graphic that illustrates the inter-relationship between entities in a database. ER diagrams often use symbols to represent three different types of information. Boxes are commonly used to represent entities. Diamonds are normally used to represent relationships and ovals are used to represent attributes. An entity–relationship model is the result of using a systematic process to describe and define a subject area of business data. It does not define business process; only visualize business data. The data is represented as components (entities) that are linked with each other by relationships that express the dependencies and requirements between them, such as: one building may be divided into zero or more apartments, but one apartment can only be located in one building. Entities may have various properties (attributes) that characterize them. Diagrams created to represent these entities, attributes, and relationships graphically are called entity–relationship diagrams.

Symbols used in ER diagram: -

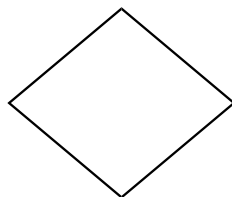
Entity:



Attributes



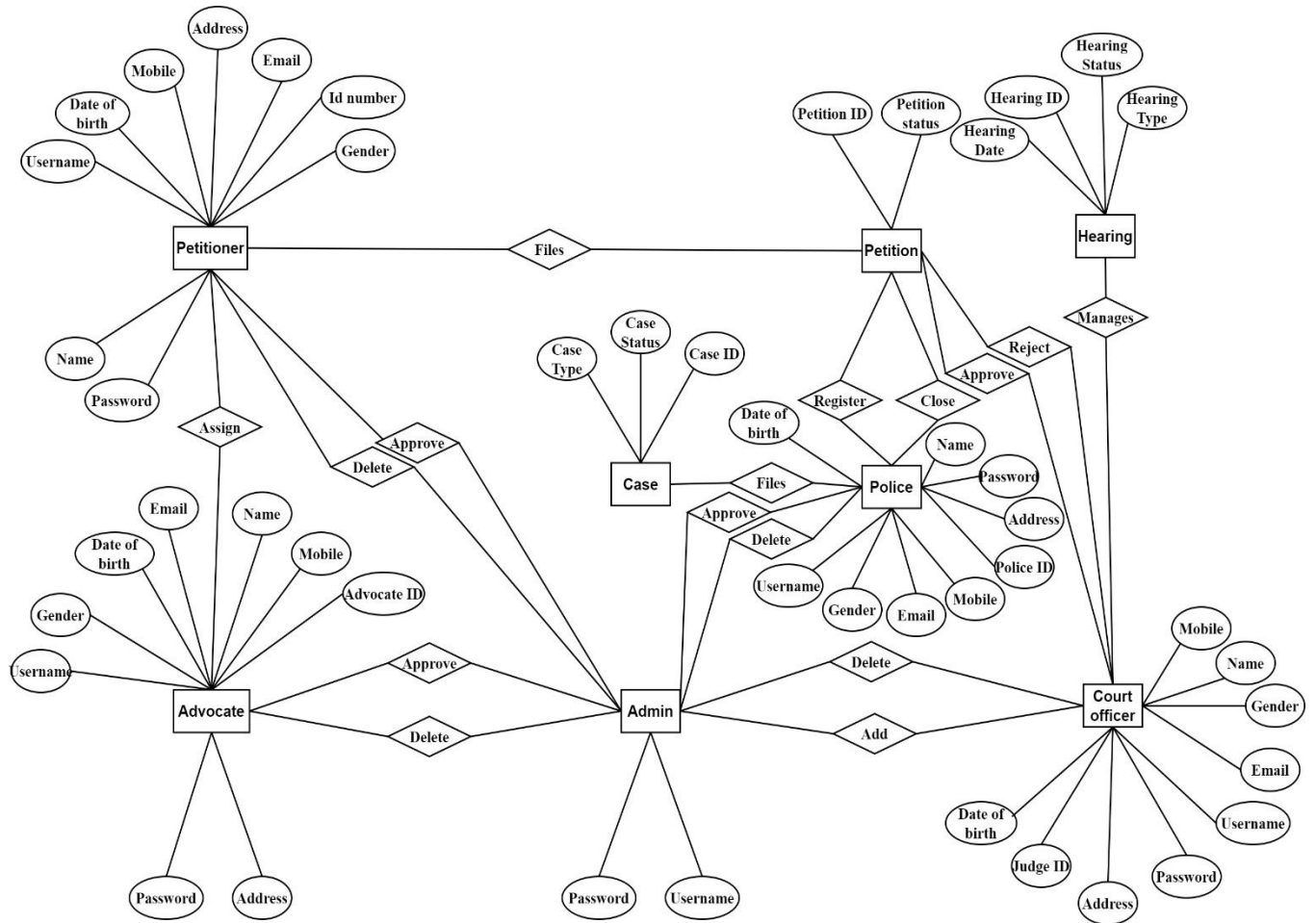
Relationship:



Lines:



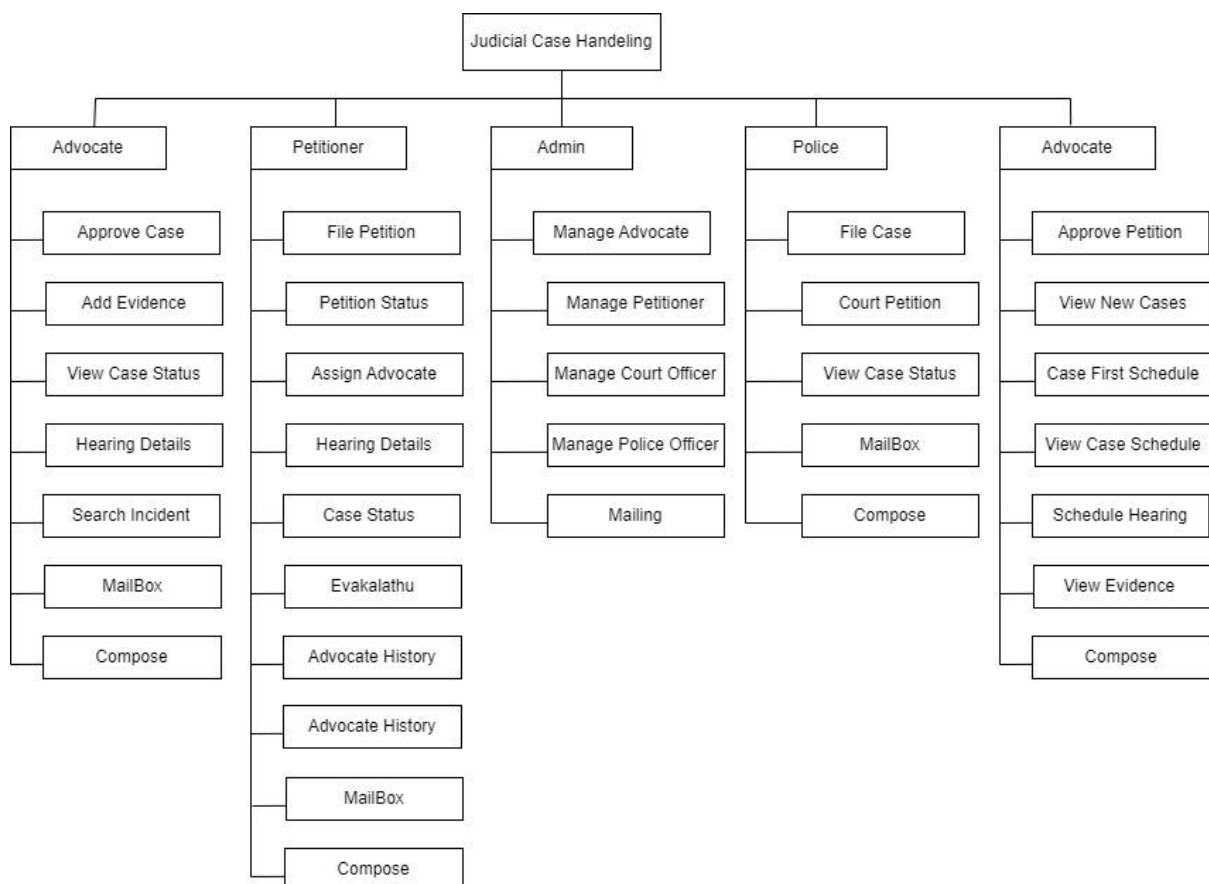
ENTITY RELATIONSHIP DIAGRAM



3.5 ARCHITECTURAL DESIGN

3.5.1 Structure Chart

A structure chart (SC) in software engineering and organizational theory, is a chart which shows the breakdown of a system to its lowest manageable levels. They are used in structure programs to arrange modules into a tree. Each module is represented by a box, which contains the module's name. The tree structure visualizes the relationships between the modules.



3.6 PROCEDURAL DESIGN

3.6.1 Flow Chart

A flowchart is a type of diagram that represents an algorithm, workflow or process, showing the steps as boxes of various kinds, and their order by connecting them with arrows. This diagrammatic representation illustrates a solution model to a given problem. Flowcharts are used in analysing, designing, documenting or managing a process or program in various fields.

Symbols used:

Flow line:



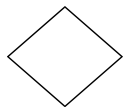
Terminal:



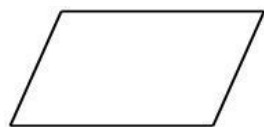
Process:



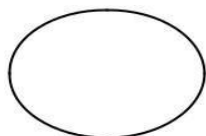
Decision:



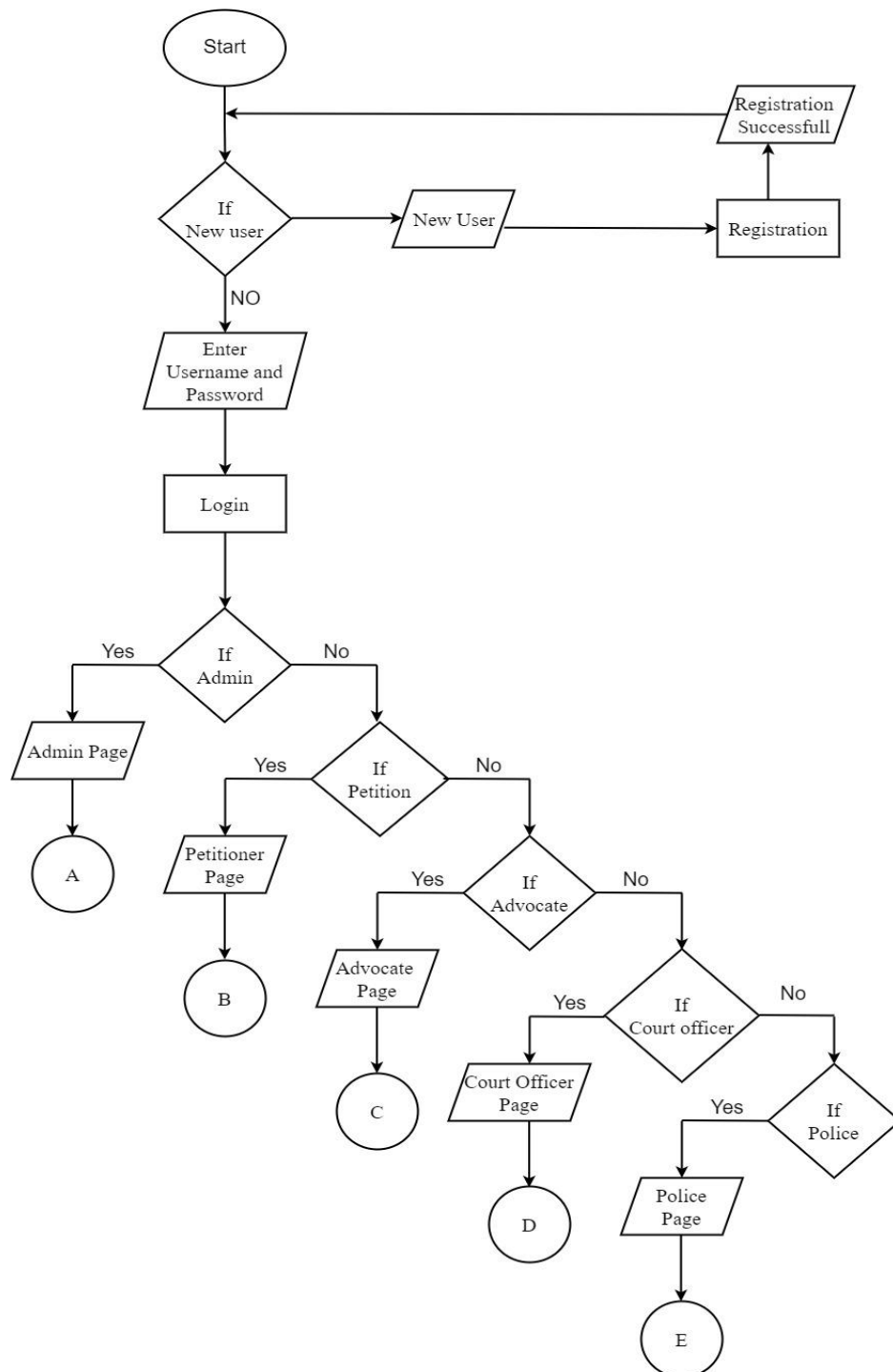
Input/Output:



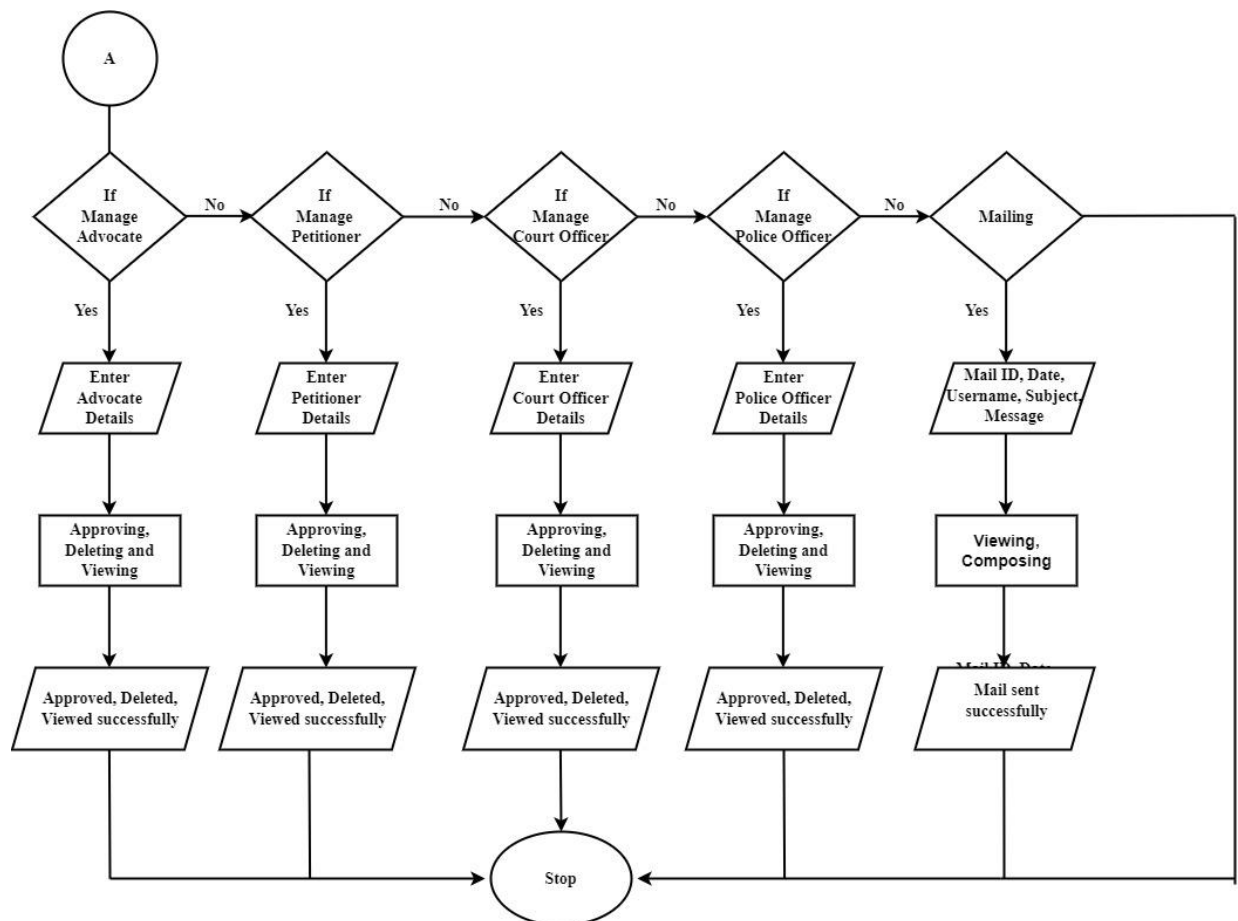
Connector:



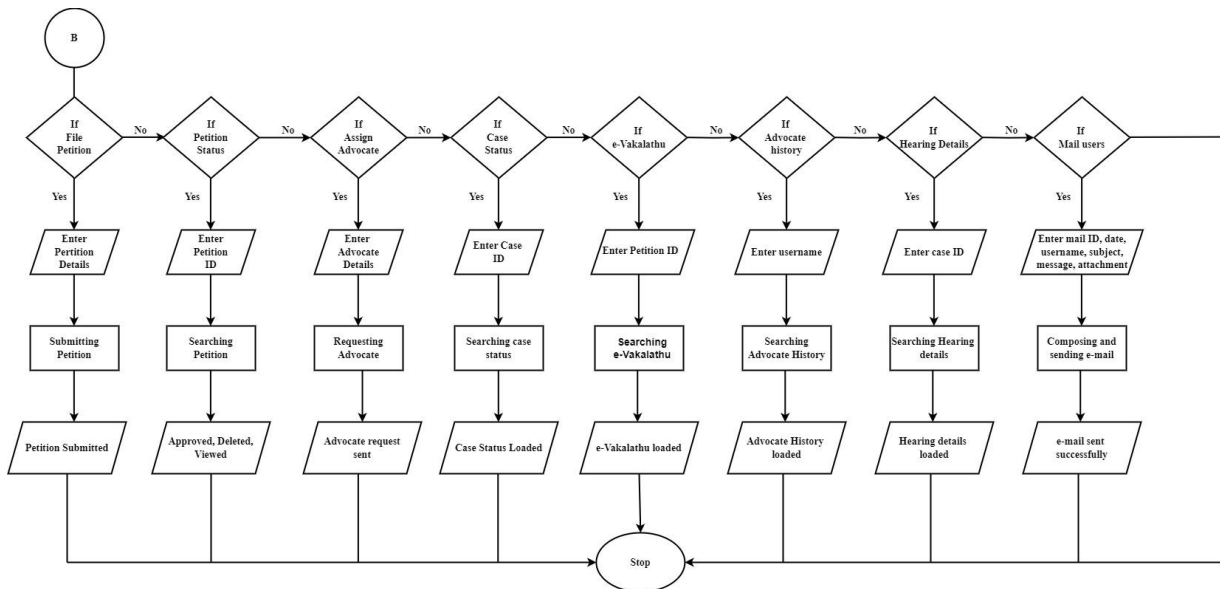
Flow Chart of Judicial Case Handling System



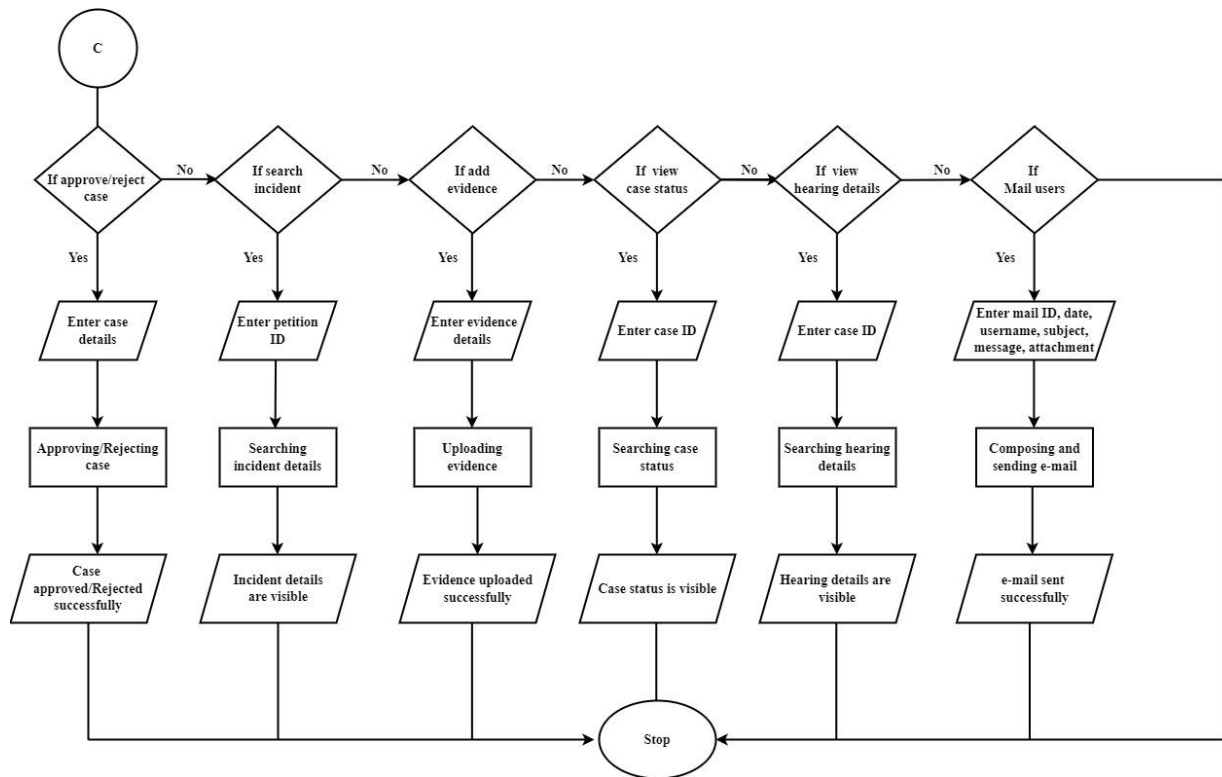
Flow chart of Admin



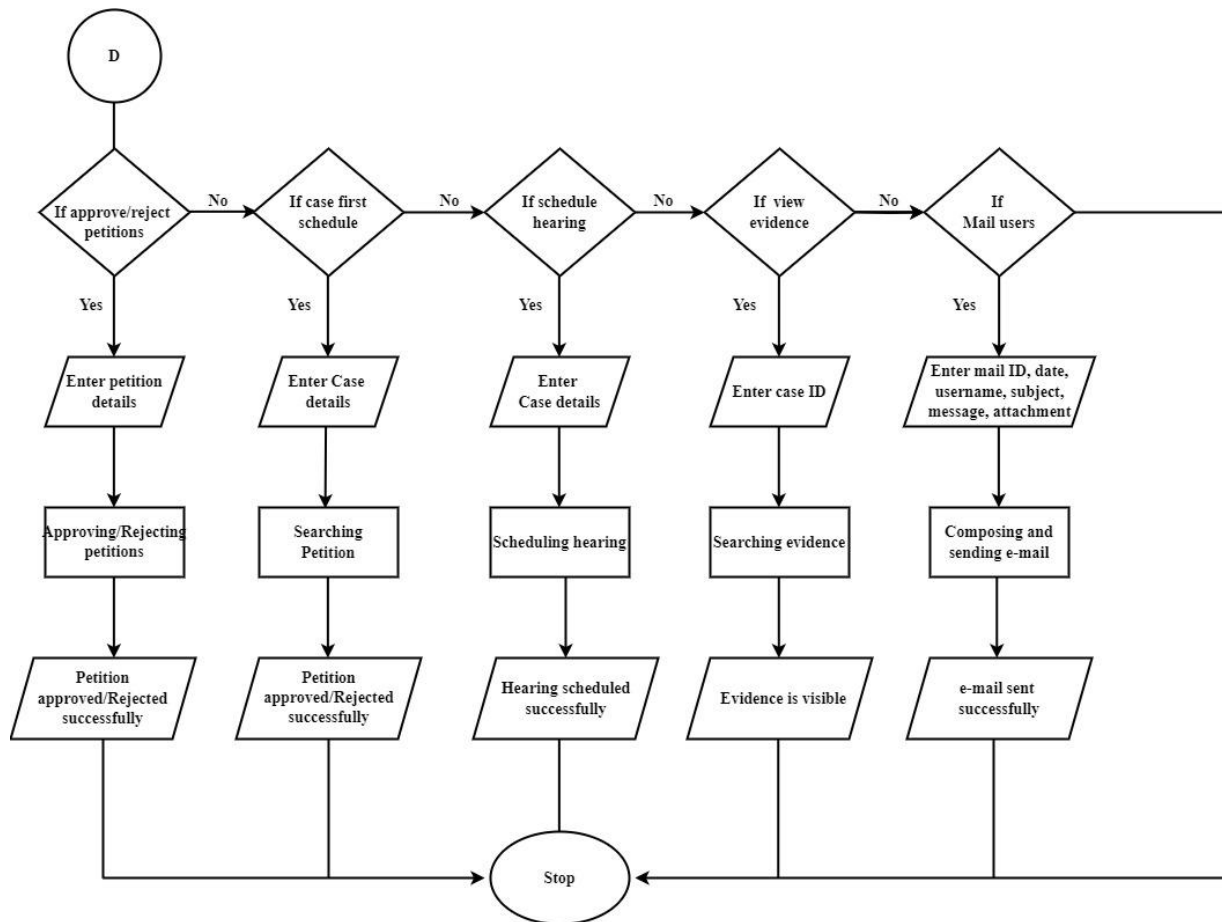
Flow chart of petitioner



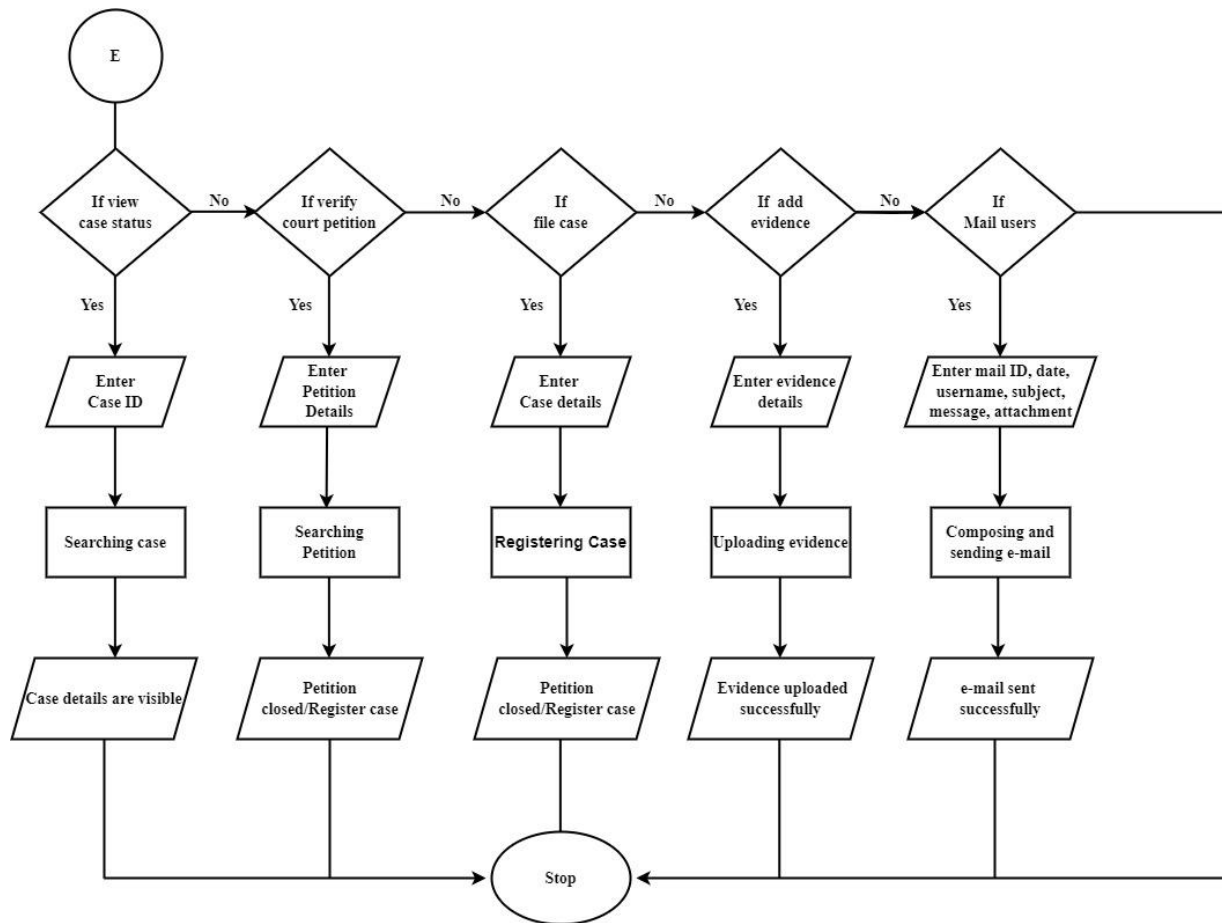
Flow chart of advocate



Flow chart of Court officer



Flow chart of police officer



PROCESSING ENVIRONMENT

4.1 HARDWARE SPECIFICATIONS

Intel core i5 Processor

An Intel Core i5 is a powerful processor for consumer-end computers and devices. It offers excellent performance and is equipped with Intel Turbo Boost Technology, similar to the Core i7. The Core i5 is available in various versions, ranging from dual-core to hexa-core, allowing for efficient multitasking. It can handle up to 6 threads simultaneously. The processor clock speed of the Core i5 ranges from 1.60 GHz to 4.90 GHz, with cache memory varying from 4 to 12 MB. The thermal design power (TDP) of the Core i5 can range from 15 watts to 125 watts. The Core i5 is well-suited for multitasking, moderate media editing, gaming, and other demanding tasks. It provides a balance between performance and affordability for users.

Processor speed

With technology advancements, increasing productivity goals, faster internet speeds, and the proliferation of devices, the need for speed has become a crucial aspect of our daily lives. We have become accustomed to instant results and expect our devices to keep up with our multitasking demands. When it comes to high-performance technology, computer processors and their clock speed are key factors we consider. The speed of a computer processor, also known as CPU speed, plays a vital role in determining the performance of a computer. The CPU is often referred to as the "brain" of the computer, and its proper functioning is essential for the longevity and functionality of the system. To understand what constitutes a good processor speed, it is important to grasp the role of the processor and how its components enhance the overall functionality of the computer.

RAM

RAM (Random Access Memory) is the hardware in a computing device where the operating system (OS), application programs and data in current use are kept so they can be quickly reached by the device's processor. RAM is the main memory in a computer. This memory stores a temporary data. Temporary data means as long as your computer system is running, your data is stored in RAM and as soon as your computer is shut down, the data stored in RAM gets deleted. RAM is volatile in nature, which means, the data is lost when the device is switched off. RAM is known as the Primary memory of the computer. RAM is known to be expensive since the memory

can be accessed directly. RAM is the fastest memory; therefore, it is an internal memory for the computer.

Hard Disk Capacity

A computer hard disk drive (HDD) is a non-volatile data storage device. Non-volatile refers to storage devices that maintain stored data when turned off. All computers need a storage device, and Hard Disk Drive are just one example of a type of storage device. HDD are usually installed inside desktop computers, mobile devices, consumer electronics and enterprise storage arrays in data centres. They can store operating systems, software programs and other files using magnetic disks.

Keyboard

A multimedia keyboard is one with media keys — additional buttons, typically along the top, for controlling audio playback, for starting common applications (e.g., e-mail client and Web browser) and other auxiliary functionality. Many such keyboards also contain a volume knob, implemented as a rotary encoder so as to be able to provide relative volume changes regardless of the volume level set by the user in the operating system and application software.

Mouse

A mouse is a small device that a computer user pushes across a desk surface in order to point to a place on a display screen and to select one or more actions to take from that position. The mouse first became a widely-used computer tool when Apple Computer made it a standard part of the Apple Macintosh.

USB

USB 2.0, also known as hi-speed USB was introduced in 2000. It is an updated version of USB 1.1, which provides improved functionalities and better speed. It is capable to deliver the maximum transfer speed of 480 Megabits per second. However, practically it is approximately 280 Mbps. USB 3.0, also known as SuperSpeed USB was first made available in November 2009. It is a much-improved version of USB 2.0. It supports the data transfer rate of 5 Gigabits per second, which is much faster than the speed provided by USB 2.0.

4.2 SOFTWARE SPECIFICATIONS

Windows 11

Windows 11 is the latest major release of Microsoft's Windows NT operating system, released in October 2021. It is a free upgrade to its predecessor, Windows 10 (2015), available for any Windows 10 devices that meet the new Windows 11 system requirements. Windows 11. A version of the Windows NT operating system.

The Microsoft .NET Framework 4.0

The Microsoft .NET Framework 4.0, released in April 2010, marked a significant advancement in the evolution of Microsoft's development platform. This iteration introduced numerous enhancements and features aimed at improving developer productivity, application performance, and security. One of the key highlights of the .NET Framework 4.0 was the introduction of the Task Parallel Library (TPL) and Parallel LINQ (PLINQ), which revolutionized parallel programming by simplifying the process of writing multithreaded and parallelizable code. This allowed developers to leverage the power of modern multicore processors more effectively, resulting in faster and more scalable applications. Additionally, .NET Framework 4.0 introduced improvements to the Common Language Runtime (CLR), including background garbage collection, which helped optimize memory management and enhance application responsiveness. Other notable features included enhancements to Windows Communication Foundation (WCF), Windows Presentation Foundation (WPF), and ASP.NET, providing developers with more robust tools and frameworks for building modern and feature-rich applications across a variety of platforms and devices. With its comprehensive set of features and improvements, the Microsoft .NET Framework 4.0 continued to solidify its position as a leading development platform for building secure, reliable, and high-performance software solutions

ASP.NET

ASP.NET, developed by Microsoft, is a versatile and powerful framework for building dynamic and scalable web applications and services. Introduced in the early 2000s, ASP.NET has evolved into a comprehensive platform that empowers developers to create robust web solutions using C#, VB.NET, or any other .NET-supported language. With its Model-ViewController (MVC) architecture and Web Forms model, ASP.NET offers flexibility in application development, catering to various project requirements and developer preferences. Its rich set of built-in features, such as authentication and authorization, session management, and caching, streamline development workflows and enhance application security and performance. Moreover, ASP.NET integrates seamlessly with other Microsoft technologies, including Azure for cloud hosting and SQL Server for database management, providing a cohesive ecosystem for end-to-end application development and deployment. With its extensive documentation, vibrant community, and continuous updates, ASP.NET remains a preferred choice for developers seeking to build scalable, enterprise-grade web applications.

C#.NET

C# .NET is a versatile and powerful programming framework developed by Microsoft. Launched in the early 2000s, it combines the robustness of the C# language with the extensive libraries and tools of the .NET platform. C# .NET is used for developing a wide range of applications, from web and desktop applications to mobile and cloud-based solutions. Known for its object-oriented approach, strong type safety, and seamless integration with the Microsoft ecosystem, C# .NET offers developers a productive environment for creating high-performance applications. With features such as garbage collection, just-in-time compilation, and built-in security, C# .NET helps developers build scalable and secure applications. The framework is supported by a comprehensive set of libraries and tools, including Visual Studio, making it an ideal choice for building enterprise-grade solutions and cross-platform applications, especially when integrated with Azure cloud services and other Microsoft technologies.

MSSQL

Microsoft SQL Server (MSSQL) is a powerful relational database management system developed by Microsoft. Launched in the 1980s, MSSQL has evolved into a robust platform

that supports various data management tasks, from storing and retrieving data to performing complex analytics and reporting. Known for its scalability, reliability, and performance, MSSQL caters to a wide range of applications, from small businesses to large enterprises. With features such as transaction management, advanced security mechanisms, and built-in business intelligence tools, MSSQL offers developers and database administrators a comprehensive solution for managing and optimizing data. Furthermore, MSSQL seamlessly integrates with other Microsoft products and technologies, including .NET Framework, Visual Studio, and Azure cloud services, providing a unified ecosystem for developing, deploying, and managing database-driven applications.

VISUAL STUDIO 2022

Visual Studio 2022 is an integrated development environment (IDE) developed by Microsoft for creating a wide range of applications. Released as the latest version in the Visual Studio series, it provides developers with a rich set of tools and features for building, testing, and deploying applications across multiple platforms, including Windows, web, mobile, and cloud. Known for its intuitive user interface, robust debugging capabilities, and powerful code editing tools, Visual Studio 2022 enhances developer productivity. It includes advanced features such as IntelliCode, live share collaboration, Git integration, and support for .NET, C++, Python, and other programming languages. Visual Studio 2022 is designed for performance and scalability, enabling developers to work efficiently on both small projects and large, complex enterprise applications. With its seamless integration with Microsoft Azure and other cloud services, Visual Studio 2022 offers a comprehensive, modern development environment for building, testing, and managing applications throughout their lifecycle.

CODING AND IMPLEMENTATION

5.1 CODING

Admin Homepage(AdminHome1.aspx)

```
<%@ Page Title="" Language="C#" MasterPageFile="~/AdminMasterPage.master"
AutoEventWireup="true" CodeFile="AdminHome1.aspx.cs" Inherits="AdminHome1" %>
```

```
<asp:Content ID="Content1" ContentPlaceHolderID="head" Runat="Server">
```

```
<link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css" rel="stylesheet">
```

```
</asp:Content>
```

```
<asp:Content ID="Content2" ContentPlaceHolderID="ContentPlaceHolder1"
Runat="Server">
```

```
<section class="content">
```

```
<!-- Small boxes (Stat box) -->
```

```
<div class="row">
```

```
<div class="col-lg-3 col-xs-6">
```

```
<!-- small box -->
```

```
<div class="small-box bg-aqua">
```

```
<div class="inner">
```

```
<h3>
```

```
<asp:Label ID="Label1" runat="server" Text="Label"></asp:Label>
```

```
</h3>
```

<p>

Total Cases

</p>

</div>

<div class="icon">

<%-- <i class="ion ion-bag"></i>--%>

</div>

<%--

More info <i class="fa fa-arrow-circle-right"></i>

--%>

</div>

</div><!-- ./col -->

<div class="col-lg-3 col-xs-6">

<!-- small box -->

<div class="small-box bg-green">

<div class="inner">

<h3>

<asp:Label ID="Label4" runat="server" Text="Label"></asp:Label>

</h3>

<p>

Total Advocates

</p>

</div>

<div class="icon">

<%-- <i class="fas fa-camera fa-xs"></i>--%>

</div>

<%--

More info <i class="fa fa-arrow-circle-right"></i>

--%>

</div>

</div><!-- ./col -->

<div class="col-lg-3 col-xs-6">

<!-- small box -->

<div class="small-box bg-yellow">

<div class="inner">

<h3>

<asp:Label ID="Label2" runat="server" Text="Label"></asp:Label>

</h3>

<p>

User Petition

</p>

</div>

<div class="icon">

```

        <%--<i class="ion ion-person-add"></i>--%>

    </div>

    <%-- <a href="#" class="small-box-footer">

        More info <i class="fa fa-arrow-circle-right"></i>

    </a>--%>

</div>

</div>

<!-- ./col -->

    <div class="col-lg-3 col-xs-6">

        <!-- small box -->

        <div class="small-box bg-red">

            <div class="inner">

                <h3>

                    <asp:Label ID="Label3" runat="server" Text="Label"></asp:Label>

                </h3>

                <p>

                    Total Hearings

                </p>

            </div>

            <div class="icon">

                <%--<i class="fas fa-envelope fa-sm"></i>--%>

```

```

    </div>

    <%-- <a href="#" class="small-box-footer">

        More info <i class="fa fa-arrow-circle-right"></i>

    </a>--%>

</div>

</div><!-- /.col -->

</div><!-- /.row -->

</section><!-- /.content -->

</asp:Content>

```

Advocate Homepage(AdvocateHome1.aspx)

```

<%@
    Page
    Title="Page
    Title"
    Language="C#"
MasterPageFile="~/AdvocateMasterPage.master"
    AutoEventWireup="true"
CodeFile="AdvocateHomepage1.aspx.cs" Inherits="AdvocateHomepage1" %>

```

```

<asp:Content ID="Content1" ContentPlaceHolderID="head" Runat="Server">

    <link
        href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-
beta3/css/all.min.css" rel="stylesheet">

</asp:Content>

<asp:Content
    ID="Content2"
    ContentPlaceHolderID="ContentPlaceHolder1"
    Runat="Server">

    <section class="content">

        <div class="row">

            <div class="col-xs-12">

```

```
<div class="box">
```

```
<div class="box-header">
```

```
<h3 class="box-title">Advocate Details</h3>
```

```
</div><!-- /.box-header -->
```

```
<div class="box-body table-responsive">
```

```
<div>
```

```
<asp:DataList ID="DataList1" runat="server" Font-Names="Verdana" Font-Size="Small"
GridLines="Both" Width="600px">
```

```
<%-- <FooterStyle BackColor="#F7DFB5" ForeColor="#8C4510" />--%>
```

```
<%--<HeaderStyle BackColor="#333333" Font-Bold="True" Font-Size="Large"
ForeColor="White" HorizontalAlign="Center" VerticalAlign="Middle" />--%>
```

```
<%-- <HeaderTemplate>
```

```
Hospital Details</HeaderTemplate>
```

```
--%>
```

```
<ItemStyle BackColor="White" ForeColor="Black" Font-Names="Verdana" Font-
Size="Small" />
```

```
<ItemTemplate>
```



```
<%-- <asp:Image ID="imgEmp" runat="server" Width="200px" Height="200px"
ImageUrl='<%# Bind("uphoto") %>' style="padding-left:40px"/><br />
```

```
<br />
```

```
<br />--%>
```

```
<div style="margin-left: 10px; margin-top: 10px">
```

```
<b>Advocate ID:</b>
```

```
<asp:Label ID="Label5" runat="server" Text='<%# Bind("adid") %>' ></asp:Label>
```

```
</div>
```

```
<br />
```

```
<div style="margin-left: 10px;">
```

```
<b>Advocate Code:</b>
```

```
<asp:Label ID="Label11" runat="server" Text='<%# Bind("adcode") %>' ></asp:Label>
```

```
</div>
```

```
<br />
```

```
<div style="margin-left: 10px;">
```

```
<b>Advocate Name:</b>
```

```
<asp:Label ID="lblCName" runat="server" Text='<%# Bind("name") %>' ></asp:Label>
```

```
</div>
```

```
<br />
```

```
<div style="margin-left: 10px;">
```

```
<b>Specialization:</b>
```

```
<asp:Label ID="Label3" runat="server" Text='<%# Bind("adtype") %>' ></asp:Label>
```

```
</div>
```

```
<br />
```

```
<div style="margin-left: 10px;">
```

```
<b>Gender:</b>
```

```
<asp:Label ID="lblName" runat="server" Text='<%# Bind("gender") %>'></asp:Label>
```

```
</div>
```

```
<br />
```

```
<div style="margin-left: 10px;">
```

```
<b> Date of Birth:</b>
```

```
<asp:Label ID="lblCity" runat="server" Text='<%# Bind("dob") %>'></asp:Label>
```

```
</div>
```

```
<br />
```

```
<div style="margin-left: 10px;">
```

```
<b>Address:</b>
```

```
<asp:Label ID="Label4" runat="server" Text='<%# Bind("address") %>'></asp:Label>
```

```
</div>
```

```
<br />
```

```

<div style="margin-left: 10px;">

    <b>Mobile No:</b>

    <asp:Label ID="lblCountry" runat="server" Text='<%# Bind("phone") %>'></asp:Label>

</div>

<br />

<div style="margin-left: 10px;">

    <b>Email:</b>

    <asp:Label ID="Label2" runat="server" Text='<%# Bind("email") %>'></asp:Label>

</div>

<br />

<div style="margin-left: 10px;">

    <b>UserName</b>

    <asp:Label ID="Label8" runat="server" Text='<%# Bind("username") %>'></asp:Label>

</div>

<br />

</ItemTemplate>

</asp:DataList>

</div>

    </div><!-- /.box-body -->

    </div><!-- /.box -->

</div>

</div>

</section><!-- /.content --></asp:Content>

```

5.2. TESTING

Testing is the process of detecting errors. Testing performs a very critical role for quality assurance and for ensuring the reliability of software. The results of testing are used later on during maintenance also. The aim of testing is often to demonstrate that a program works by showing that it has no errors. The basic purpose of testing phase is to detect the errors that may be present in the program. Hence one should not start testing with the intent of showing that a program works, but the intent should be to show that a program doesn't work. Testing is the process of executing a program with the intent of finding errors.

5.2.1 Unit Testing

Unit testing plays a critical role in our Judicial case handling system, focusing on verifying the functionality of individual components or modules within the platform. Each module within Judicial case handling system, such as the Admin, Petitioner, Advocate, Court officer, and Police can be considered as a separate unit for testing. During the development phase, rigorous unit testing is performed to ensure that each module operates correctly and without errors. Inputs received from users, including registration data, product listings, and communication messages, undergo thorough validation to ensure accuracy and compliance with system requirements. By conducting unit testing on individual modules, developers can detect and address errors before they propagate to other parts of the platform. This approach allows for the identification and resolution of any issues arising from interactions between different modules, ensuring smooth and error-free functionality throughout Judicial case handling system.

Black Box Testing: In the black box testing, test cases are designed from the examination of the input/output values only and no knowledge of design or code is required. The main approaches are:

- Equivalent class partitioning
- Boundary value analysis

White Box Testing: White box testing is an important type of unit testing. A large number of white box testing strategies exist. White box testing strategies are:

- Fault based testing
- Coverage Based testing

5.2.2 Integration Testing

Integration testing is a type of software testing in which the different units, modules or components of a software application are tested as a combined entity. However, these modules may be coded by different programmers. The aim of integration testing is to test the interfaces between the modules and expose any defects that may arise when these components are integrated and need to interact with each other.

Types of test cases

Integration test case: Here we execute test cases which just tell about the connectivity from one module to other module and integrating one application to other application and how the application moves from parent node to child no demand vice versa.

Functional Test case: Here we execute test cases which tell about the functionality of the application and talks about the desired output to be seen. Internally we have different type of test cases when we write here like range, output values, BVA, ECP and so on. We give input and expect some output according to the SRS. Here we check individual module completely by checking each and every tab, text box, and buttons and so on.

Non-Functional Test case: Test cases related to user friendliness like font, image, color, how easy to navigate etc., performance related, security related comes under here.

User Acceptance test case: These test cases are crucial and very important to client-side people, because these test case talks about these business and approach of the application to complete a particular client task, which is also called as End to End Business scenario test case.

Here we won't be doing testing related to UI, Functional or Non-Functional, we talk about business and scenario for which the application is made for.

5.2.3 System Testing

Here the entire software system is tested. The reference document for this process is the requirements document, and the goal is to see if software meets its requirements. The entire software has been tested against requirements of project and it is checked whether all requirements of project have been satisfied or not.

Alpha Test: The first test of newly developed, when the first round of bugs has been fixed, the product goes with actual users for testing. For custom software the customer may be invited into the vendor's facilities for an alpha test to ensure the client's vision has been interpreted properly by the developer.

Beta Test: A test of new or revised software application that is performed by users at their facilities under normal operating conditions. Beta testing follows alpha testing. Vendors of packaged software often offer their customers the opportunity of beta testing new releases or versions and the beta testing of elaborate products such as operating systems can take months.

Acceptance testing: Testing is performed by the Client of the application to determine whether the application is developed as per the requirements specified by him/her. It is performed within the development of the organization or at the client site.

In our project the developer uses unit testing to ensure the overall working of the system.

SECURITY, BACKUP AND RECOVERY MECHANISM

Security

In addition to simple username and password authentication, the application should enforce secure password policies, such as requiring a minimum length and a combination of letters, numbers, and special characters. It should also implement measures such as encryption for sensitive user data, secure transmission of data over the network using HTTPS, and protection against common web vulnerabilities like SQL injection and cross-site scripting (XSS). Regular security audits and updates should be conducted to identify and patch any potential security vulnerabilities.

Backup

Apart from maintaining a backup source code, regular backups of the application's database and user data should be performed to ensure data integrity and availability in case of system failures or data loss incidents. A robust backup strategy should include regular automated backups, off-site storage of backup data, and periodic testing of backup restoration processes to verify their effectiveness. Additionally, users should be provided with tools and guidelines for backing up their own data stored within the application to prevent data loss due to accidental deletion or device failure.

Recovery

In addition to refreshing the application or system, the application should implement error handling mechanisms to gracefully handle unexpected errors and exceptions, providing informative error messages to users and logging detailed error information for troubleshooting purposes. Furthermore, the application should incorporate automated monitoring and alerting systems to promptly detect and respond to any issues or failures, minimizing downtime and ensuring continuous availability. In the event of a major system failure or data corruption, a comprehensive disaster recovery plan should be in place, outlining steps for restoring the application and data from backups and minimizing the impact on users. Regular testing and rehearsal of the disaster recovery plan are essential to ensure its effectiveness in real-world scenarios.

FUTURE ENHANCEMENT

As Judicial case handling system continues to evolve, there are several key enhancements that can be implemented to further improve its functionality, usability, and impact.

- Comprehensive automation of the judicial system: The e-Judiciary project aims to fully automate the judicial process.
- Integration with other governmental systems: Cross-agency collaboration can link with tax record databases, land registries, and public safety information. This would allow seamless access to relevant data during investigations and court proceedings.
- Real-time data exchange: Real-time data sharing between the judiciary and enforcement agencies enhances coordination and case access.
- User experience improvement: Developing a mobile application for users like judges, lawyers, petitioners, and police officers allows on-the-go access and real-time notifications. Continuous improvement of the user interface based on feedback ensures accessibility for all technical abilities.
- Artificial Intelligence (AI) and Machine Learning: Predictive analytics can forecast case outcomes, aiding in better decision-making for lawyers and judges. Automated document review systems can identify errors and inconsistencies in legal documents, improving case preparation.
- Blockchain exploration: Blockchain ensures integrity with tamper-proof court records and case documentation. Smart contracts automate legal processes, reducing manual intervention and improving efficiency.
- Public engagement and transparency: Public dashboards with case resolution times and pending cases enhance transparency. User feedback mechanisms enable continuous system improvement based on stakeholders' experiences.
- Training and support for users: Comprehensive training programs will ensure users understand the system. Dedicated support teams will provide one-on-one assistance and technical support for ongoing user engagement.

SOFTWARE MAINTAINANCE

Software maintenance is widely accepted part of SDLC now days. It stands for all the modifications and update done after the delivery of software product.

Software Maintenance planning includes ten activities:

- **Preparation** – Describe software preparation and transition activities including the conception and creation of the maintenance plan; describe how to handle problems identified during development and configuration management.
- **Modification** – After the application has become the responsibility of the maintenance team, explain how to analyse each request; confirm and check validity; investigate and propose solutions; document the proposal and get the required authorizations to apply the modifications.
- **Implementation** – Describe the process for considering the implementation of the modification itself.
- **Acceptance** – Describe how the modification is accepted by the maintenance team.
- **Migration** – Describe any migration tasks that need to be executed. If the software needs to be moved to another system, outline the steps to do so without impacting its functionality.
- **Transition** – Document the sequence of activities to transition the system from Development to Maintenance.
- **Service Level Agreements** – Document SLAs and maintenance contracts negotiated by Maintenance.
- **Change Request** – Outline the problem-handling process to prioritize, documents and route change and maintenance requests.
- **Modification Request acceptance/reject**– Describe the request including details of the size/effort/complexity. If this is too complex to resolve, outline the steps to route the issue back to the software team.
- **Retirement** – This is the final stage in the lifecycle. Describe how to retire the software and the steps to archive any data that may be a by-product of the system.

Types of Maintenance

Traditionally, 5 types of maintenance have been distinguished, which are differentiated by the nature of the tasks that they include:

Corrective maintenance: The set of tasks is destined to correct the defects to be found in the different equipment and that are communicated to the maintenance department by users of the same equipment.

Preventive Maintenance: Its mission is to maintain a level of certain service on equipment, programming the interventions of their vulnerabilities in the most opportune time. It is used to be a systematic character, that is, the equipment is inspected even if it has not given any symptoms of having a problem.

Predictive Maintenance: It pursues constantly know and report the status and operational capacity of the installations by knowing the values of certain variables, which represent such state and operational ability. To apply this maintenance, it is necessary to identify physical variables (temperature, vibration, power consumption, etc.) and to understand which type of variation is indicative of problems that may be appearing on the equipment. This maintenance it is the most technical, since it requires advanced technical resources, and at times of strong mathematical, physical and / or technical knowledge.

Zero Hours Maintenance (Overhaul): The set of tasks whose goal is to review the equipment at scheduled intervals before appearing any failure, either when the reliability of the equipment has decreased considerably so it is risky to make forecasts of production capacity. This review is based on leaving the equipment to zero hours of operation, that is, as if the equipment were new. These reviews will replace or repair all items subject to wear. The aim is to ensure, with high probability, a good working time fixed in advance.

Periodic maintenance (Time Based Maintenance TBM): The basic maintenance of equipment made by the users of it. It consists of a series of elementary tasks (data collections, visual inspections, cleaning, and lubrication, retightening screws) for which no extensive training is necessary, but perhaps on brief training. This type of maintenance is the based on TPM (Total Productive Maintenance).

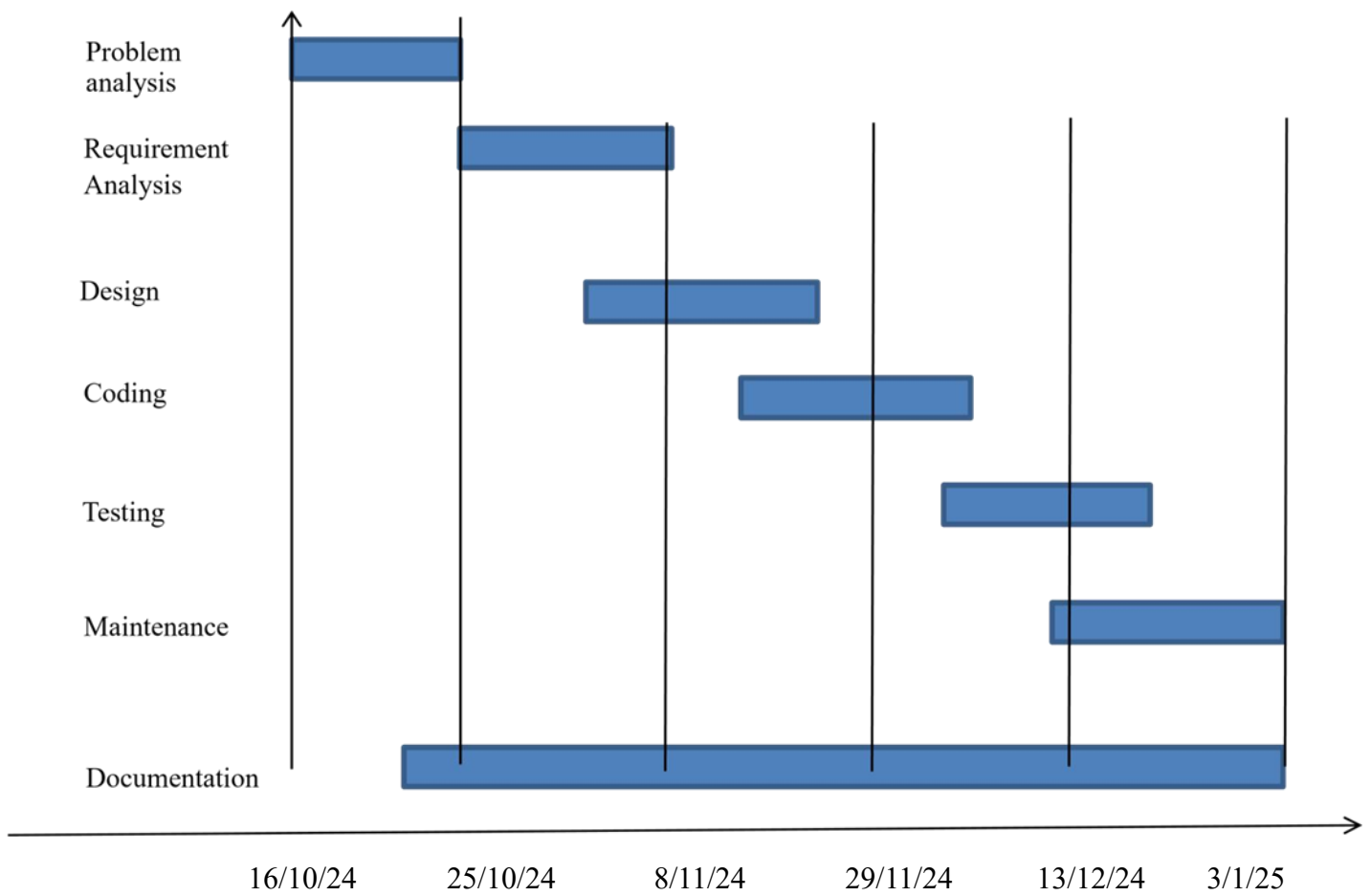
CONCLUSION

The Judicial case handling system is a revolutionary model of change transforming the judicial landscape into a more efficient, transparent, and accessible mechanism for all stakeholders involved with it. The project takes help from advanced technologies and innovative approaches to overcome longstanding challenges in the judiciary, including delays, lack of accessibility, and administrative inefficiencies. The proposed modules of judges, lawyers, petitioners, police officers, and court administrators create a comprehensive ecosystem that streamlines processes and fosters collaboration among various entities.

This system will be further enhanced in the future with its integration to the government databases, mobile applications, AI-driven analytics, as well as blockchain security. e-Judiciary promises to further enhance the capabilities of this system and works for priority of user experience and access, as well. Citizen and legal professions will be empowered equally toward obtaining equal access to all judicial resources and information. In the long run, this approach strengthens not just the rule of law but also inspires confidence in the judicial process for creating a far-fairer society in which justice is not only done but seen to be done. The dream of an entirely automated judiciary is a huge leap toward the prospects of a modern, sensitive legal system that is responsive to everyone's needs, irrespective of their social background.

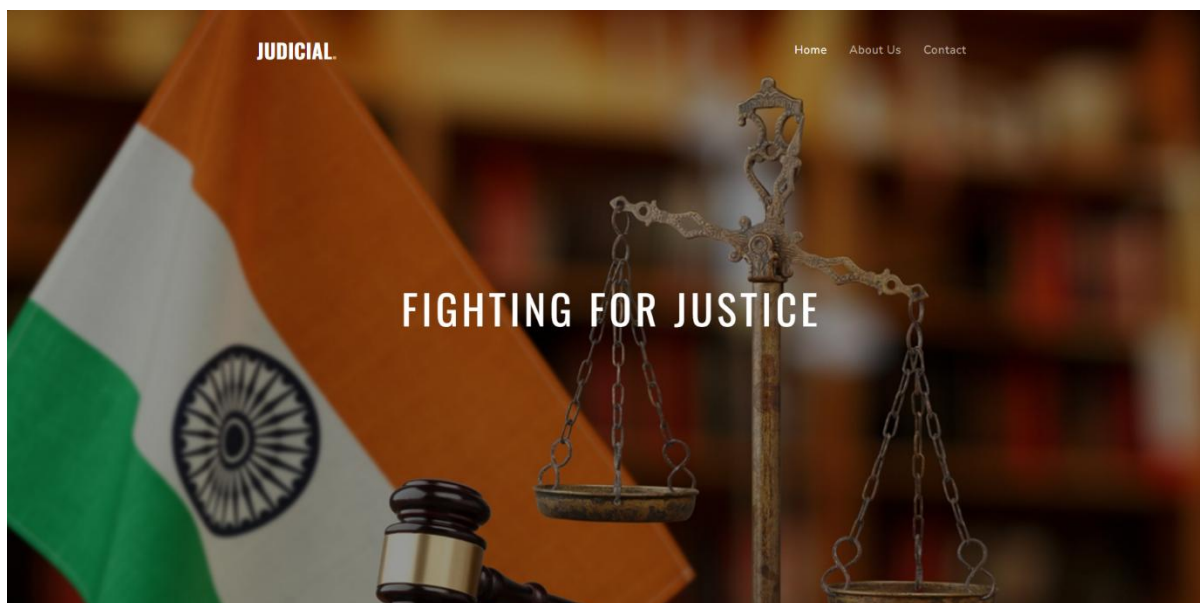
APPENDIX

GANTT CHART

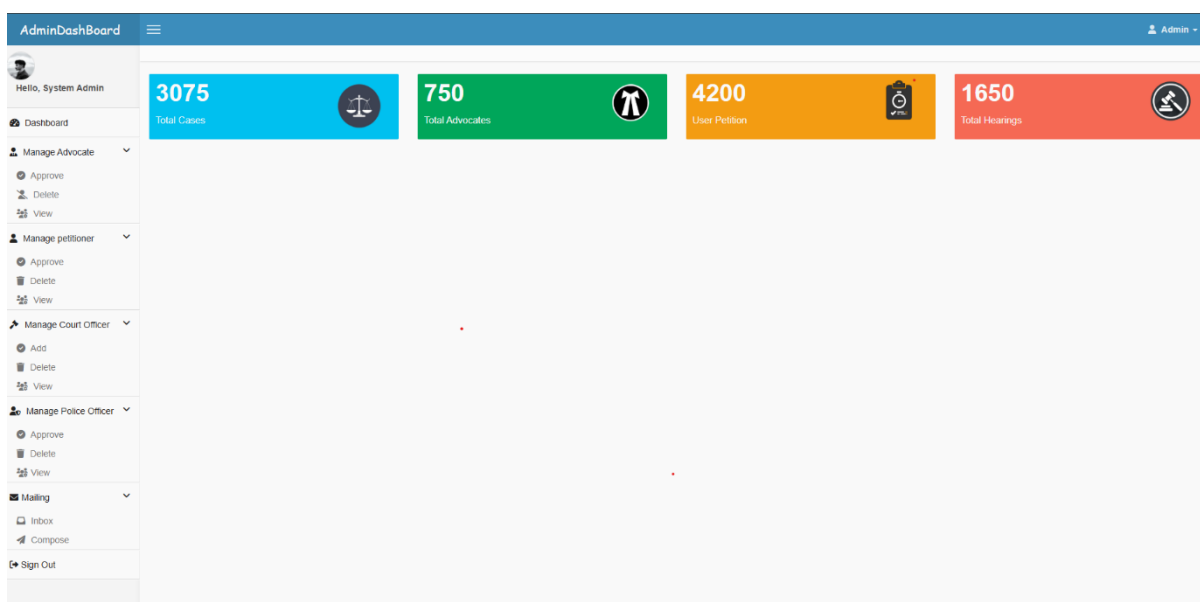


SCREENSHOTS

Homepage



Admin Dashboard



Petitioner Registration Page


Home Petitioner Registration Advocate Registration Police Registration

Police Registration

A officer is a law enforcement professional who works in courtrooms to enforce laws and courtroom regulations.

Advocate Registration

An advocate is a professional that speaks and advocates for a particular category of person, such as a customer, a victim, a child or a patient.

Petitioner Registration

A petitioner is a person who pleads with governmental institution for a legal remedy or a redress of grievances, through use of a petition.

Petitioner Registration

Username: Gender:

Name: Contact No:

Email ID: DOB:

Password: ID Proof Type:

Confirm Password: ID Proof No:

Address: Petitioner ID:

Petitioner Homepage

Petitioner Dashboard

- HomePage
- File Petition
- Petition Status
- Assign Advocate
- Case Status
- Evakalathu
- Advocate History
- Hearing Details
- MailBox
- Compose
- Sign Out

Petitioner ID: 201

Petitioner ID Proof: HND123456

Petitioner Name: raju

Gender: MALE

Date of Birth: 15/10/1960

Address: sreekariyam

Mobile No: 9847206938

Email: raju@gmail.com

Username: raju123

Petitioner Advocate History

Petitioner: DashBoard

HomePage

File Petition

Petition Status

Assign Advocate

Case Status

Evakalathu

Advocate History

Hearing Details

MailBox

Compose

Sign Out

Advocate History

Advocate

unni123

Case Details

Case ID	Case Type	IPC Section	Case No	Police Station	Petitioner Username	Case Date	Incident Date	Location	Attachment
508	ARBITRATION PETITION	IPC 999	508/38/2025	FORT STATION	raju123	07-02-2025	23-01-2025	Pattom	Download

Personal Details

Advocate ID	BAR Code	Name	Phone	Email	Gender	Username	Court Reg	Photo
103	ID124794	unni	9964827368	unni@gmail.com	MALE	unni123	19/12/1964	

Police Add Case

Police DashBoard

HomePage

File Case

Court Petition

View Case Status

MailBox

Compose

Sign Out

Petition ID

422

Case ID:

542

Case No:

542/70/2025

Case Type:

SELECT

Police Station:

FORT STATION

Petition Date:

24-02-2025 20:31:05

Register Date:

11-03-2025

IPC Section:

Petition Username:

raju123

Incident Area:

lvm

Respondent Username:

SELECT

Incident Date:

24-2-2025

Court Officer:

anoop123

Details:

accident

Register Case

Close

Police View Case Status

Police DashBoard

HomePage

File Case

Court Petition

View Case Status

MailBox

Compose

Sign Out

Police View Case Details

Case ID

531

Search

Case Details

Case ID	Case Type	IPC Section	firmo	Police Station	Petitioner Username	Case Date	Incident Date	Location	Attachment
531	ARBITRATION PETITION	IPC 123	531/56/2025	FORT STATION	raju123	25-02-2025	02-02-2025	maranalloor	Download

Evidence Details

Evidence ID	Case ID	firmo	Petitioner Username	Petitioner Advocate	Evidence Type	Evidence Date	Attachment
809	531	531/56/2025	raju123	mahesh123	image	25-02-2025	Download

Hearing Details

Hearing ID	Case ID	firmo	Petitioner Advocate	Respondent uname	Respondent Advocate	Hearing Date	Hearing Type	Judgement Details	Case Status
717	531	531/56/2025	mahesh123	chikku123	0	03-03-2025	offline	0	Closed

Court Officer Approve Petition

Court Officer

Hello, Court Officer

Search...

Approve Petition

View New Cases

Case First Schedule

View Case Schedule

Schedule Hearing

View Evidence

View Case Status

Sign Out

Court Officer Approve Petition

Incident ID

455

Incident ID

455

Petition Type

ARBITRATION PETITION

UserName

01-03-2025 12:10:33

Police Station

FORT STATION

Petition Date

raju123

Incident Date

accident

Incident Area

mki

Incident Details

accident

Download

[Download](#)

[Allocated](#)

[Reject](#)

Admin View Police

AdminDashBoard

Admin

Hello, System Admin

Dashboard

Manage Advocate

- Approve
- Delete
- View

Manage petitioner

- Approve
- Delete
- View

Manage Court Officer

- Add
- Delete
- View

Manage Police Officer

- Approve
- Delete
- View

Mailing

- Inbox
- Compose

Sign Out

Admin View Police

Police ID	Police Code	Name	Police Station	Date of Birth	Mobile	Email	Gender	Username
301	IN123485	Anoop	FORT STATION	16/7/1983	9847206938	anoop@gmail.com	MALE	anoop123
303	HND173456	niclu	KAZHAKUTTOM	18/2/1994	9847297648	niclu@gmail.com	MALE	niclu123
308	ajdyap	Ajchandran	Thiruvallam	9/6/1976	9447044756	ajchandran@gmail.com	MALE	Ajchandran123

Admin Mail Compose

Hello, System Admin

Dashboard

Manage Advocate

- Approve
- Delete
- View

Manage petitioner

- Approve
- Delete
- View

Manage Court Officer

- Add
- Delete
- View

Manage Police Officer

- Approve
- Delete
- View

Mailing

- Inbox
- Compose

Sign Out

Mail ID: 606

Date: 20-03-2025 19:05:08

To: rajur123

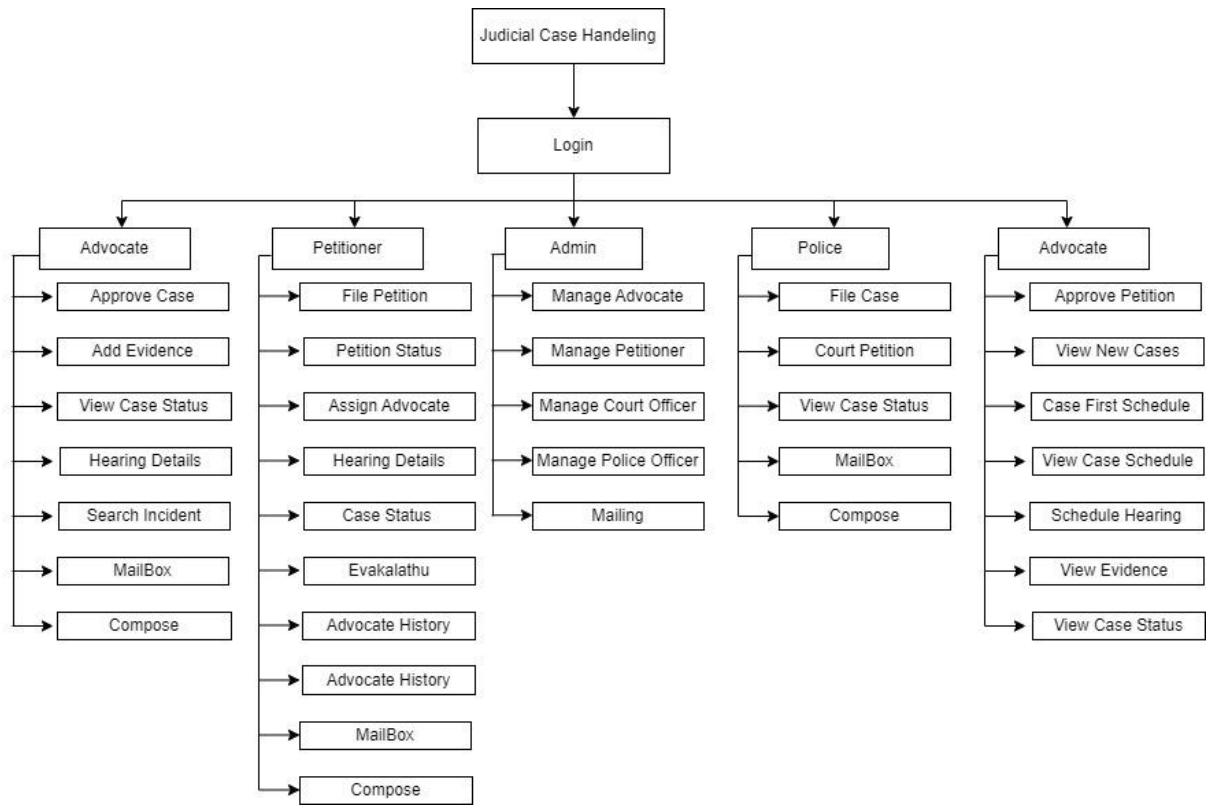
Subject: test

Attachment: Choose File SET 3 (1).pdf

Message: This is a test message

Send

MENU TREE



MEETING MINUTES

DATE: 16/10/2024

Place: Bethel Computers Malayam

Discussion: Discussed about the topic

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 08/11/2024

Place: Bethel Computers Malayam.

Discussion: Came up with e judiciary

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 15/11/2024

Place: Bethel Computers Malayam.

Discussion: Researchers made on the existing system

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 22/11/2024

Place: Bethel Computers Malayam.

Discussion: Discussion on various modules and functionalities

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 29/11/2024

Place: Bethel Computers Malayam.

Discussion: Started C#.NET introduction class

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 06/12/2024

Place: Bethel Computers Malayam.

Discussion: Started design of the Login Page

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 13/12/2024

Place: Bethel Computers Malayam.

Discussion: Continued designing

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 20/12/2024

Place: Bethel Computers Malayam.

Discussion: Documentation class

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 03/01/2025

Place: Bethel Computers Malayam.

Discussion: Design for the project was prepared.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 07/01/2025

Place: Christ Nagar College Maranalloor

Discussion: Internal guide review.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 10/01/2025

Place: Bethel Computers Malayam.

Discussion: Coding started for admin module

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 17/01/2025

Place: Bethel Computers Malayam.

Discussion: Coding for admin module continued.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 21/01/2025

Place: Christ Nagar College Maranalloor

Discussion: Internal guide review.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 24/01/2025

Place: Bethel Computers Malayam.

Discussion: Coding for petitioner, Court officer module started.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 31/01/2025

Place: Bethel Computers Malayam

Discussion: Coding Petitioner, court officer module continued.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 07/02/2025

Place: Bethel Computers Malayam.

Discussion: Coding for police module started.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 14/02/2025

Place: Bethel Computers Malayam

Discussion: Coding for advocate module started.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 21/01/2025

Place: Christ Nagar College Maranalloor

Discussion: Internal guide review.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 21/02/2025

Place: Bethel Computers Malayam.

Discussion: Designing of ER Diagram.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 28/02/2025

Place: Bethel Computers Malayam.

Discussion: Designing of Flowchart

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 21/01/2025

Place: Christ Nagar College Maranalloor

Discussion: Internal guide review.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 07/03/2025

Place: Bethel Computers Malayam.

Discussion: Designing of Flowchart

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 14/03/2025

Place: Bethel Computers Malayam.

Discussion: Completion of updating

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

DATE: 21/01/2025

Place: Christ Nagar College Maranalloor

Discussion: Internal guide review.

Members: Abhinav Chandra A R, Nikhil Sajan, Sooraj V, Devanandan A Nair

BIBLIOGRAPHY

Book References

1. Roger S Pressman, *"Software Engineering"*, McGraw Hill Education, 6th Edition.
2. Tanweer Alam, *"Web Designing and development"*, 1st Edition, Khanna Book Publishing, 2014.
3. Penny Grubb, *"Software Maintenance Concepts and Practice"*, 2nd Edition.
4. Rajib Mall, *"Fundamentals of software engineering"*, 5th Edition, PHI Learning, 2018.

Website References

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2. <https://www.krishi-market.com/>
3. <https://agrowala.com/>
4. <https://geeksforgeeks.com/>
5. <https://amazon.in/>